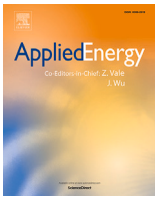
Applied Energy 388 (2025) 125637



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen





Review of machine learning techniques for optimal power flow☆

Hooman Khaloie*, Mihály Dolányi, Jean-François Toubeau, François Vallée

Power Systems and Markets Research Group, Electrical Power Engineering Unit, University of Mons, Mons, 7000, Belgium

HIGHLIGHTS

- Offers in-depth overview and classification of OPF learning models, addressing past review gaps.
- Comapres learning models by OPF type, dataset and network size, metrics, algorithms, etc.
- Presents generalizable metrics for learning-based OPF, creating a performance evaluation framework.
- Pinpoints challenges at the intersection of OPF and learning methods, guiding future research.

ARTICLE INFO

Keywords:
End-to-end (E2E) learning
Learning-to-optimize (L2O)
Machine learning
Optimal power flow (OPF)
Power systems

ABSTRACT

The Optimal Power Flow (OPF) problem is the cornerstone of power systems operations, providing generators' most economical dispatch for power demands by fulfilling technical and physical constraints across the power network. To ensure safe and reliable operation of power systems, grid operators must steadily solve the nonconvex nonlinear OPF problem for immense power networks in (near) real-time, which poses tremendous computational challenges. The enormous amount of available data created by power systems digitalization and recent breakthroughs in machine learning have opened up new opportunities for grid operators to build shortcuts to predict or solve the OPF problem close to real-time. This survey overviews recent attempts at leveraging machine learning algorithms to solve the transmission-level OPF problem. On this basis, the groundwork is laid for commonly employed machine learning approaches leveraged to address the OPF problem. Subsequently, the frequently used performance evaluation metrics in learning-based OPFs are delineated to judge efficiency from diverse aspects (e.g., optimality in terms of the dispatched cost, feasibility concerning technical constraints, and computational efficiency) compared to conventional approaches. Next, the trend and progress of recently developed algorithms are discussed. Finally, the challenges and open problems at the interface of machine learning and OPF problems are highlighted.

1. Introduction

Power systems operations rely on the Optimal Power Flow (OPF) problem, which determines generators' most cost-effective dispatch to satisfy power demands while accounting for technical and physical constraints throughout the power network [1]. The Optimal Power Flow (OPF) problem serves as the cornerstone for a range of power system studies, addressing both short-term concerns such as unit commitment [2], security-constrained economic dispatch [3], market clearing [4], transmission network congestion management [5], voltage control [6], and optimal transmission switching [7], as well as long-term considerations like transmission expansion planning [8]. Through extensive research, scholars have sought to find fast and efficient solutions to the nonconvex nonlinear Alternating Current OPF (AC-OPF) problem for many years. Centered on short-term considerations, finding a quick and effective solution to the OPF problem is of utmost importance for grid operators, particularly given the heightened fluctuations in power supply and demand. Routinely, the OPF problem is solved every five to fifteen minutes by grid operators to maintain the secure and steady operation of the power network [9]. Solving this computationally expensive problem with hundreds/thousands of decision variables frequently for large-scale power networks causes grid operators to often resort to simplified or approximated versions of the original OPF problem.

☆ This document is the results of the research project of the Energy Transition Fund, DISCRETE project, funded by the FPS Economy, S.M.E.s, Self-Employed and Energy, Belgium.

* Corresponding author.
Email address: hoomankhaloiee@gmail.com; hooman.khaloie@umons.ac.be (H. Khaloie).

https://doi.org/10.1016/j.apenergy.2025.125637
Received 27 May 2024; Received in revised form 23 January 2025; Accepted 23 February 2025
Available online 7 March 2025
0306-2619/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Abbreviations

AC-OPF	Alternating current optimal power flow
ADMM	Alternating direction method of multipliers
DC-OPF	Direct current optimal power flow
DDPG	Deep Deterministic Policy Gradient
E2E	End-to-end
GNN	Graph neural network
IPOPT	Interior point optimizer
KKT	Karush–Kuhn–Tucker
L2O	Learning-to-optimize
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MSE	Mean square error
NN	Neural network
NRMSE	Normalized root mean square error
OPF	Optimal power flow
PPO	Proximal policy optimization
RES	Renewable energy source
RMSE	Root mean square error
SAC	Soft actor-critic
TD3	Twin delayed deep deterministic policy gradient

With an eye on the above, the following main challenges revolve around the OPF problem [10–12].

- Given the rising levels of uncertainty and stochasticity in both demand and supply sides resulting from intermittent Renewable Energy Sources (RESs) and demand response schemes, it is imperative to adjust resource setpoints with even greater frequency to ensure compatibility with the ever-shifting system states, which is restricted by the computationally expensive nature of the OPF problem. Faced with this challenge, grid operators are often pressed into service to exercise simplified or approximated variants of the AC-OPF problem, such as Direct Current OPF (DC-OPF), which is less computationally expensive. Adopting such a strategy typically has multiple consequences. Firstly, these oversimplified models yield suboptimal or infeasible solutions due to overlooking the inherent complexities of the power system. Secondly, omitting or oversimplifying fundamental network constraints, as in DC-OPF, may result in unrealistic solutions and compromised system security.
- To maintain the secure and safe operation of power systems, grid operators commonly explore security-constrained OPF, which is a stochastic variant of the classic OPF. This involves envisioning various line and generator contingencies, each introducing additional power flow constraints to the main optimization problem, resulting in exponentially high computational complexity.
- Large electricity grids, such as the United States grid, consist of numerous interconnected regions. Each region is overseen by a system operator who possesses comprehensive information about their designated area but lacks access to details from other regions, such as market bids, network configurations, and system states [13]. To preserve security and privacy among various regions, the electricity grid should be dispatched in a decentralized manner. Decentralized OPF can be realized via diverse techniques, most notably the Alternating Direction Method of Multipliers (ADMM), which may result in numerical instability and convergence issues related to the tuning parameters [14,15], even variants with the convergence guarantees [15,16].

A consensus has arisen recently regarding the potential of data-driven machine learning methods in addressing the aforementioned challenges. These methods enable the representation of highly intricate relationships and enable swift prediction of variables. The main idea of machine learning methods is to transfer the computational overhead from online optimization to offline training processes by utilizing extensive historical data [17]. Historical data can be gathered through various instruments such as supervisory control and data acquisition[1] systems [18], phasor measurement units[2] [19], smart meters [20], fault recorders [21], or utilizing solutions encompassing a wide range of scenarios representing different power system states. In the context of the OPF, machine learning methods seek to utilize abundant historical data to train an offline process, enabling *rapid direct solution prediction* or *acceleration of traditional OPF optimizers*. These two overarching pathways are commonly known as End-to-End (E2E) learning and Learning-to-Optimize (L2O) approaches. In the E2E learning approach, OPF results are directly derived from given inputs without explicitly solving the OPF optimization problem. Conversely, in the L2O approaches, learning algorithms support traditional optimizers in the decision-making process to obtain OPF results. These two pathways have garnered significant attention recently, with a wide range of evolving machine learning techniques.

Although the application of machine learning methods in different domains of electric and power system studies has been reviewed in several works [22–26], the existing literature surveys on machine learning-based OPFs are summarized into an informative tutorial [27], a concise survey [28], and several other generic reviews [24,29,30] covering the OPF problem as a subsidiary issue. The tutorial in [27] provided an informative overview of E2E and L2O approaches for the OPF problem [28], focusing on the mathematical foundations and a step-by-step implementation guide. In the previous survey on the application of machine learning for the OPF problem [28], the authors attempted to capture the trend by categorizing the literature into six distinct directions: direct prediction of the OPF results, prediction of the active set of constraints, prediction of binary variables, learning the control policy, data-driven stability-constrained OPF, and prediction of warm-start points. While this work offers valuable insights into OPF learning schemes, given the presently growing rate of developed methodologies, the lack of a comprehensive review of machine learning approaches for the OPF problem with a well-laid classification of recent endeavors is a research gap that has not been filled in previous works [27,28]. Overall, this review makes the following contributions:

1. Provides a comprehensive overview and classification of OPF learning schemes to address the lack of a comprehensive review in previous works. The presented OPF learning classification categorizes L2O approaches into three groups: *learning active sets of constraints*, *warm-start*, and *learning-assisted distributed/simplified optimization*. Simultaneously, E2E learning models undergo a primary classification into *supervised mapping*, *unsupervised mapping*, and *reinforcement learning*. Subsequently, these E2E learning models are subjected to a secondary taxonomy involving categorization into *unconstrained* or *constrained* and/or *graph learning* techniques.
2. Conducts a thorough comparative analysis of OPF learning models considering factors such as OPF type, network size, evaluation metrics, input parameters, dataset variations, dataset size, security analysis, learning algorithm, and open-source code availability.

[1] SCADA
[2] PMUs

For the first time, this work conducts a comparative analysis of OPF learning studies to explore the factors mentioned above.
3. Presents an extensive range of common evaluation metrics for assessing the performance of OPF learning schemes, establishing a standardized framework for performance evaluation.
4. Identifies challenges and open research directions at the intersection of machine learning and OPF, guiding future studies and advancements in the field.

The remaining sections are structured as follows. Section 2 presents an overview of OPF learning schemes, which includes a brief introduction to the OPF problem, an overview of representative machine learning-driven OPF schemes, and a discussion of common evaluation metrics used in OPF schemes. Sections 3 and 4 present a comparative analysis of E2E and L2O schemes for the OPF problem, respectively. Section 5 discusses the challenges and open issues with respect to OPF learning models, and finally, Section 6 concludes the paper.

2. Overview of learning-based OPF schemes

In this section, a succinct overview of OPF learning schemes is presented. The OPF problem is introduced first. Subsequently, representative OPF learning architectures are introduced, with key features highlighted.

2.1. Optimal power flow problem

The primary objective of the OPF problem is to determine the optimal allocation of generators' resources, considering a range of technical and physical constraints existing in the power network, in order to satisfy power demands while minimizing costs. The OPF problem, in a simplified form that omits security constraints and renewable generation, can be expressed as follows [27]:

$$\underset{\Xi}{\text{Min}} \sum_{b\in\mathcal{B}} f_b^{\text{cost}}(p_b^g) \tag{1a}$$

subject to:

$$p_{b,n}^f = g_{b,n}v_b^2 - v_b v_n(b_{b,n}\sin(\theta_b-\theta_n) + g_{b,n}\cos(\theta_b-\theta_n)) \quad \forall (b,n)\in\mathcal{L} \tag{1b}$$

$$q_{b,n}^f = -b_{b,n}v_b^2 - v_b v_n(g_{b,n}\sin(\theta_b-\theta_n) - b_{b,n}\cos(\theta_b-\theta_n)) \quad \forall (b,n)\in\mathcal{L} \tag{1c}$$

$$p_b^g - p_b^d = \sum_{(b,n)\in\mathcal{L}} p_{b,n}^f \quad \forall b\in\mathcal{B} \tag{1d}$$

$$q_b^g - q_b^d = \sum_{(b,n)\in\mathcal{L}} q_{b,n}^f \quad \forall b\in\mathcal{B} \tag{1e}$$

$$\underline{v}_b \le v_b \le \overline{v}_b \quad \forall b\in\mathcal{B} \tag{1f}$$

$$\underline{\theta}_{b,n} \le \theta_b - \theta_n \le \overline{\theta}_{b,n} \quad \forall (b,n)\in\mathcal{L} \tag{1g}$$

$$\underline{p}_b^g \le p_b^g \le \overline{p}_b^g \quad \forall b\in\mathcal{B} \tag{1h}$$

$$\underline{q}_b^g \le q_b^g \le \overline{q}_b^g \quad \forall b\in\mathcal{B} \tag{1i}$$

$$(p_{b,n}^f)^2 + (q_{b,n}^f)^2 \le (\overline{s}_{b,n}^f)^2 \quad \forall (b,n)\in\mathcal{L} \tag{1j}$$

where the goal of (1a) is to identify the active power dispatch p_b^g at bus b that minimizes the overall operational cost $f_b^{\text{cost}}(\cdot)$ while adhering to equality (1b)–(1e) and inequality (1f)–(1j) constraints. The system models a power network consisting of buses $b/n \in \mathcal{B}$ and transmission lines (b,n) in the set $\mathcal{L}$. The active and reactive power produced by generators are represented by p_b^g and q_b^g, while the line flows are denoted by $p_{b,n}^f$ and $q_{b,n}^f$. Voltage magnitude and angle at each bus are represented by v_b and θ_b. These variables jointly form the set of OPF decision variables, $\Xi = \{p_b^g, q_b^g, p_{b,n}^f, q_{b,n}^f, v_b, \theta_b\}$. In contrast, the OPF receives the load values p_b^d and q_b^d as inputs, along with conductance $g_{b,n}$ and susceptance $b_{b,n}$ for each transmission line. It is noteworthy that while minimizing the cost of generators' dispatch is a common approach, alternative objectives, such as minimizing differences between nodal voltage values and their nominal levels, may also be considered based on the specific goals and constraints of the analysis. In (1), the equality constraints (1b)–(1e) encompass power flow and power balance equations. The operational constraints defined by (1f)–(1j) limit voltage magnitudes, differences in phase angles, and the levels of active and reactive power from generators, while also controlling the power flows through transmission lines. It is important to note that the formulation of new power flow equations continues to be an ongoing research topic. Readers can explore other types of power flow models by referring to the monograph [12].

The nonlinearity inherent within power flow equations, along with additional operational constraints reshapes the feasible space of the OPF problem into a non-convex domain, thereby creating a formidable NP[3]-hard optimization challenge. This complexity is heightened by introducing discrete decision variables, exemplified by generators' on/off status. On the whole, two prominent pathways are conventionally pursued to address this NP-hard optimization [12]:

1. Utilizing mathematical approximation or relaxation techniques to effectively address the inherent nonlinear elements within the problem, thereby yielding a more streamlined model.
2. Implementing nonlinear programming algorithms, such as the Interior point optimizer (IPOPT), aimed at achieving local optimum.

While the first approach may produce infeasible solutions in terms of power flow equations, the second approach guarantees feasibility. Although the computational intensity of these power flow solution methods poses challenges for real-life networks, it is crucial to note that the issue extends beyond the solution time. As the complexity of the problem increases, methods like IPOPT may yield suboptimal solutions or fail to converge to an optimum. Additionally, when employing relaxations, the accuracy diminishes with growing complexity, increasing the likelihood of encountering infeasible or less feasible solutions to the original problem. For a thorough examination of traditional mathematical approximation and relaxation techniques applied in addressing the OPF problem and a comprehensive formulation of the problem itself, readers are encouraged to refer to [12]. On the contrary, Sections 2.2 and 2.3 offer alternative approaches designed to tackle the NP-hard OPF optimization using machine learning and data-driven techniques.

2.2. Representative OPF learning schemes

To alleviate the computational challenges posed by the OPF problem, data-driven machine learning methods have garnered recognition as technically effective solutions. Learning-based schemes for solving the OPF problem can be divided into two main groups, E2E learning and L2O, as introduced in Fig. 1 and Table 1.

2.2.1. E2E learning schemes

E2E learning schemes for the OPF problem aim to derive the OPF solution (e.g., the output power of generators) directly from given inputs (e.g., load and RESs). Instead of relying on traditional optimization approaches outlined in 2.1, these schemes provide a *direct mapping* from inputs to the OPF solution, thereby offering substantial computational advantages. E2E schemes involve a two-step process. Initially, a machine learning model learns the intricate relationship between input data and OPF outcomes using extensive historical data in an offline procedure. Subsequently, during real-time operations, the model predicts the OPF solution for a given set of input data.

E2E learning approaches for OPF can be systematically classified into primary and secondary categories based on the learning mechanism and

[3] Non-deterministic Polynomial-time

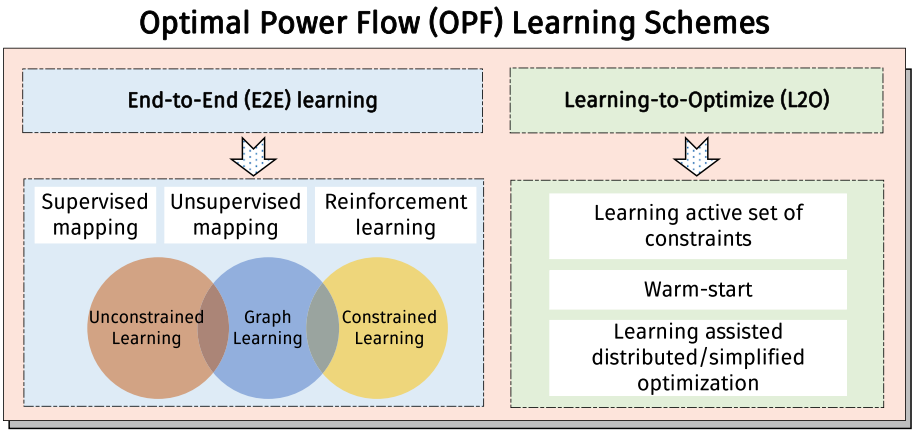


Fig. 1. Overview of OPF learning schemes.

Table 1
Representative OPF learning schemes.

E2E Learning Schemes	**Description**	Learn a mapping from input features to output without explicitly formulating the OPF problem		
	Advantages	Ability to capture complex relationships within power system data, high computational efficiency		
	Disadvantages	(Mostly): Lack of interpretability, inability to generalize rare system states		
	Representative classes	***Supervised Mapping:*** Learn the mapping from input features to OPF solution using labeled training data	***Unsupervised Mapping:*** Limited solely to generative models, learn the distribution of OPF decisions	***Reinforcement Learning:*** Train an agent to interact with the power system environment through trial-and-error to find an OPF solution
		The above main classes can be further classified as *unconstrained* or *constrained* and/or *graph learning* models		
	Representative subclasses	***Unconstrained Learning:*** Learn the mapping without incorporating operational constraints in learning process	***Constrained Learning:*** Learn the mapping incorporating selected/all operational constraints in learning process	***Graph Learning:*** Learn the mapping leveraging graph representations of power systems
L2O Schemes	**Description**	Learn algorithms/strategies to efficiently aid OPF optimization process		
	Advantages	Higher level of interpretability, leveraging advantages of machine learning and optimization simultaneously		
	Disadvantages	Possible challenges in generalizing to rare OPF instances not well-represented in training data		
	Representative classes	***Learning Active Set of Constraints:*** Learn to predict the active set of constraints for a given OPF instance	***Warm-Start:*** Leverage pre-trained models to provide initial feasible solutions/starting points for OPF solvers	***Learning-assisted distributed/simplified optimization:*** Combine distributed/simplified optimization techniques with learning models

the type of dataset employed, as detailed in Table 1. The primary class incorporates well-established E2E learning schemes, specifically *supervised mapping* [31], *unsupervised mapping* [32], and *reinforcement learning* [33]. These primary classes can be further categorized into subclasses such as *unconstrained* **or** *constrained* [34], **and/or** *graph learning* [35]. Each subclass provides additional insights into primary OPF learning models, which offer distinctions in terms of considering or disregarding operational constraints and/or representing power systems as graph-based data. The following briefly overviews these E2E OPF learning approaches.

In the primary class of E2E learning schemes, the OPF problem, depending on the type of dataset (labeled or unlabeled), can be framed as a mapping from input features to OPF solutions by utilizing either supervised or unsupervised learning models. In supervised mapping, the OPF problem is tackled by training a machine learning model on a labeled dataset. This dataset contains explicit pairings between input data (such as load profiles) and corresponding OPF results (e.g., generators' output power and voltage magnitudes). By learning from this labeled dataset, the model becomes capable of predicting OPF results for new, unseen input data, such as unseen load profiles.

The OPF problem is traditionally framed as a supervised learning task, unlike the unsupervised learning models commonly employed for clustering, dimensionality reduction, and association tasks [32]. The exploration of unsupervised mapping models within the OPF domain appears to be largely limited, to the best of the authors' knowledge, to generative adversarial networks [36]. This implies a notable departure from the conventional supervised approach and highlights the limited adoption of unsupervised learning models in the OPF landscape, where they solely learn from unlabeled data. Interestingly, the generative adversarial networks model the distribution of OPF decisions and generate corresponding OPF solutions for new input datasets. These networks have two primary modules: the *generator*, responsible for producing OPF solutions, and the *discriminator*, tasked with determining whether these solutions are genuine or artificial. By aiming to mislead the discriminator, the generator adjusts its outputs to align with the underlying data patterns. Over time, the generator improves its ability to create realistic and valid OPF predictions. This zero-sum interaction, where one module's success is the other's failure, defines the unsupervised nature of the learning framework. Unlike supervised learning, the goal is not to reduce the gap between predictions and labeled ground truth but to successfully deceive the discriminator. Consequently, the approach is referred to as unsupervised, as it prioritizes misleading the discriminator rather than minimizing error relative to labeled data. Further details on the application of generative adversarial networks in learning OPF solutions are provided in Section 3.2.

Differently, reinforcement learning standardizes the OPF problem as a sequential decision-making process within a Markov decision process framework. In this framework, OPF solutions are obtained by training an

agent, which acts as the power system operator, to learn a policy through iterative interactions with the environment, typically represented by a power network simulator. By trial-and-error, the agent explores actions, observes resulting states and rewards, and gradually learns to make decisions leading to favorable OPF outcomes.

The aforementioned primary E2E learning models can be subcategorized into classes labeled *unconstrained* **or** *constrained* [34], **and/or** *graph learning* [35]. Each main class (supervised, unsupervised, and reinforcement) can be characterized as either an unconstrained, constrained, or graph learning approach. Nevertheless, the latter (graph learning) may overlap with either the unconstrained or constrained learning paradigms, as illustrated in Fig. 1. With these in mind, constrained learning aims to incorporate the operational constraints of the OPF problem, such as line flow limits, into the learning process to ensure the feasibility of the obtained solutions. This approach typically utilizes various techniques, including Lagrange multipliers, penalty methods, and implicit layers, to integrate operational constraints within the learning procedure [37–39]. These techniques, widely interpreted as physics-aware or physics-informed learning [24], provide a means to address the inherent physical and operational constraints of the OPF problem within an E2E learning framework, leading to more reliable and practical results. On the other hand, unconstrained models learn the mapping from inputs to the OPF solution by disregarding the operational constraints during the learning procedure. Note that unconstrained learning approaches may yield infeasible solutions, which necessitate post-processing steps for adapting their feasibility in relation to the physical and operational constraints of the power system. Lastly, graph learning leverages graph representations and techniques to integrate power systems topology into the learning process to deal with ever-changing grid topology under contingencies.

2.2.2. L2O schemes

The second category of learning schemes for solving the OPF problem centers on L2O frameworks. These frameworks aim to leverage learning algorithms or strategies that effectively support traditional OPF optimization models in solving the problem. L2O frameworks are also referred to as *indirect*, *hybrid machine learning and optimization*, or *learning-assisted optimization* models in the literature. While data-driven machine learning models offer numerous ways to assist traditional optimization procedures, the existing research on L2O schemes for the OPF problem can be broadly classified into three classes: *Learning active set of constraints*, *warm-start*, and *learning-assisted distributed/simplified optimization*.

Learning an active set of constraints as a class of L2O centers on eliminating non-binding inequality constraints of the OPF problem, i.e., those constraints that are not active at the optimal operating point, through machine learning models. It is widely acknowledged that the high computational expense associated with the OPF problem can be significantly alleviated by eliminating a substantial number of inactive inequality (non-binding) constraints [40], thereby yielding a streamlined optimization problem that offers cost-effective solutions. Alternatively, another line of research within L2O is to leverage supervised or reinforcement learning techniques to generate a solution, although not feasible but relatively close to the optimal solution, to warm-start OPF solvers. By warm-starting the OPF solvers with a solution close to the global optimal point, standard solvers will require fewer iterations (less computational cost) to converge and attain the optimal solution. Finally, learning-assisted distributed/simplified optimization models concentrate on utilizing learning models to support optimization algorithms developed for *i)* approximation and relaxation of the OPF and *ii)* dispatching the power system in a decentralized/distributed manner. An example of this class of L2O is enhancing the convergence speed of the ADMM, a well-established distributed optimization approach, through machine learning models.

2.3. Performance evaluation metrics in learning-based OPF

When it comes to fully or partially substituting traditional optimization approaches with learning schemes, the primary concern is how well these alternatives perform from various perspectives [41]. Particularly centering on the OPF, various evaluation metrics are widely exploited to judge the performance of learning models against a consensual benchmark such as IPOPT. These evaluation metrics offer comprehensive insights into the strengths and weaknesses of the employed approach. By assessing various performance aspects, these metrics enable a detailed analysis of effectiveness, objective measurement, and evaluation of trade-offs associated with the approach. For machine learning-driven OPF, a wide range of evaluation metrics is available, which can vary depending on the specific learning framework utilized. In this overview, the common performance evaluation metrics for OPF learning are generally classified into two categories: *solution quality* and *computational efficiency and convergence*.

2.3.1. Solution quality

Solution quality metrics cover three distinct aspects to evaluate the effectiveness or quality of the solutions derived from learning schemes. Firstly, they appraise optimality by evaluating how well the learning approach returns the objective function value, i.e., minimum dispatch cost (as given in (1a)). Secondly, solution quality metrics gauge feasibility by exploring the scheme's ability to satisfy OPF operational constraints, which encompass both equality (1b)–(1e) and inequality (1f)–(1j) constraints (e.g., limits on voltage and power variables). Lastly, statistical measures are employed to evaluate the accuracy and predictive performance of various variables derived from learning algorithms (e.g., active power of generators) compared to the actual values (ground truth). These solution quality metrics comprehensively evaluate the capability of OPF learning models to deliver optimal solutions, maintain feasibility, and accurately predict relevant variables. The following gives specifics on these metrics.

1. **Optimality**:
   As a well-documented metric, optimality refers to the degree of proximity between the objective function (e.g., dispatch cost) obtained by a learning scheme and the dispatch cost generated by a widely accepted benchmark solver (ground truth), such as IPOPT. To calculate the optimality, the first step is to compute the optimality loss (gap), which is determined by evaluating the Mean Absolute Percentage Error (MAPE) of the objective function across a set of T test samples, as expressed in Eq. (2). Subsequently, the optimality is computed based on Eq. (3).

$$\text{Optimality loss } (\%) = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{f\left(P_t^{\circ}\right)-f\left(P_t^{*}\right)}{f\left(P_t^{*}\right)}\right| \tag{2}$$

$$\text{Optimality } (\%) = 100\ \% - \text{Optimality loss } (\%) \tag{3}$$

   where t is the index of test samples and T represents the total number of test samples. The variables P_t° and P_t^{*} represent the set of generators' output power derived from a learning scheme and a benchmark solver, respectively. Similarly, $f\left(P_t^{\circ}\right)$ and $f\left(P_t^{*}\right)$ respectively denote the dispatch costs associated with the machine learning model and the benchmark solver. The optimality is quantified on a percentage scale, with higher values indicating superior performance.

2. **Feasibility**:
   Feasibility, as one of the utmost concerns of OPF learning frameworks, measures how well the leveraged method can respect

the OPF operational constraints. Although the majority of machine learning models exhibit rapid OPF solution retrieval (once trained), it is crucial to evaluate the applicability of the obtained solution considering technical limitations (e.g., line flow constraints). Within the relevant literature, feasibility is quantified by measuring constraint violation and/or constraint satisfaction rate along with the feasibility rate, as introduced in the following.

(1) *Constraint Violation*:
Constraint violation is characterized as a deviation from the intended operational state of the power system, which includes both OPF inequality and equality constraints. Quantifying this deviation depends on the chosen perspective and can involve diverse approaches, such as utilizing various norms (L_1, L_2, L_∞) and calculating the maximum or average constraint violation across T test samples. In terms of the former, constraint violation can be defined as follows:

$$\lambda = \frac{1}{T}\sum_{t=1}^{T} \|\theta_t\|_{1/2/\infty} \tag{4}$$

where θ_t records constraint violation at instance t. Assuming that the variable x derived from a learning model in test instance t is represented by x_t°, the constraint violation θ_t for any arbitrary inequality constraint, $\underline{X} \le x \le \overline{X}$, is defined as follows:

$$\theta_t = \max\left\{\underline{X} - x_t^\circ, 0\right\} + \max\left\{x_t^\circ - \overline{X}, 0\right\} \tag{5}$$

where $\underline{X}$ and $\overline{X}$ are the allowed lower and upper bounds of the intended variable x, respectively. While Eq. (5) provides the precise measure of the constraint violation, it might be more appropriate to utilize normalized constraint violation $\check{\theta}_t$ for a sensible scaling through the following equation:

$$\check{\theta}_t = \max\left\{\frac{\underline{X} - x^\circ}{|\overline{X} - \underline{X}|}, 0\right\} + \max\left\{\frac{x_t^\circ - \overline{X}}{|\overline{X} - \underline{X}|}, 0\right\} \tag{6}$$

This normalized value represents the relative violation of the constraint. Given that the learning model generates the values x_t° and y_t° for variables x and y in sample t, the calculation of constraint violation θ_t for a given equality constraint in the form of $x + y = z$ is performed as follows:

$$\theta_t = |x_t^\circ + y_t^\circ - z| \tag{7}$$

To calculate the normalized constraint violation for a given equality constraint, the exact value in Eq. (7) is divided by the fixed value (input parameter) specified in the equality constraint (assume it is denoted by z):

$$\check{\theta}_t = \frac{|x_t^\circ + y_t^\circ - z|}{|z|} \tag{8}$$

The scale of quantifying constraint violations in Eqs. (5) and (7) is consistent with the scale of the involved variables, while the scale in the normalized values (6) and (8) is expressed per unit.

(2) *Feasibility Rate and Constraint Satisfaction Rate*:
Alternatively, the quality of solutions obtained by learning techniques can be assessed by measuring the feasibility and constraint satisfaction rates. These metrics are defined as:

$$\text{Feasibility rate } (\%) = \frac{n_f}{n_r} \times 100\ \% \tag{9}$$

$$\text{Constraint satisfaction rate } (\%) = \frac{n_s}{n_c} \times 100\ \% \tag{10}$$

where n_f, n_r, n_s, n_c denote the total number of feasible solutions, the total number of solutions, the total number of satisfied equality and inequality constraints, and total number of constraints, respectively.

3. **Statistical Measures**:
Statistical measures play a pivotal role in evaluating the effectiveness of machine learning models, which offer valuable insight into the accuracy and predictive abilities of these frameworks [42]. By facilitating the quantification of the discrepancy between predicted and actual values of the system's state, these measures create a foundation for assessing the model's efficacy and making informed decisions about the performance. Assuming that x_t° and x_t^* are the solutions obtained through a machine learning framework (predicted value) and a consensual solver (actual value or ground truth) for an arbitrary variable x in test sample $t \in T$, the following provides an overview of the statistical measures frequently employed in OPF learning studies.

(1) *Relative Error*:
Relative error quantifies the magnitude of difference (error) between predicted and actual values relative to the actual value in percentage and is defined by:

$$\textit{Relative Error}(\%) = \left|\frac{x_t^\circ - x_t^*}{x_t^*}\right| \times 100\ \% \tag{11}$$

(2) *Mean Absolute Error (MAE)*:
The MAE, by keeping the same scale as the output variable, represents the average magnitude of the errors between predicted and actual values and is calculated by:

$$MAE = \frac{1}{T}\sum_{t=1}^{T} |x_t^\circ - x_t^*| \tag{12}$$

(3) *Mean Absolute Percentage Error (MAPE)*:
The MAPE expresses the average percentage error between predicted and actual values in percentage and is determined by:

$$MAPE(\%) = \frac{1}{T}\sum_{t=1}^{T} \left|\frac{x_t^\circ - x_t^*}{x_t^*}\right| \times 100\ \% \tag{13}$$

(4) *Mean Square Error (MSE)*:
The MSE, by placing more emphasis on larger errors, quantifies the average squared error between predicted and actual values, as in (14):

$$MSE = \frac{1}{T}\sum_{t=1}^{T} \left(x_t^\circ - x_t^*\right)^2 \tag{14}$$

It is important to acknowledge that the MSE does not retain the original unit of the predicted variable.

(5) *Root Mean Square Error (RMSE)*:
The RMSE, by keeping the same scale as the output variable, quantifies the differences between predicted values and actual values, squaring the errors, taking the mean, and then finding the square root. The RMSE is calculated as follows:

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \left(x_t^\circ - x_t^*\right)^2} \tag{15}$$

(6) *Normalized Root Mean Square Error (NRMSE)*:
The NRMSE, derived from the RMSE, standardizes errors by dividing them by an appropriate metric related to the variable, facilitating comparisons between various variables or datasets. Generally, RMSE normalization can be approached by dividing RMSE by mean, standard deviation, difference between maximum and minimum predicted values, and interquartile range, as denoted in (16) [43].

$$NRMSE = \begin{cases} \frac{RMSE}{\frac{1}{T}\sum_{t=1}^{T} x_t^{\circ}} & \rightarrow \text{Mean} \\ \frac{RMSE}{\mathrm{SD}(x_t^{\circ})} & \rightarrow \text{Standard deviation} \\ \frac{RMSE}{\max(x_t^{\circ})-\min(x_t^{\circ})} & \rightarrow \text{Difference between maximum and minimum} \\ \frac{RMSE}{Q_3-Q_1} & \rightarrow \text{Interquartile range} \end{cases} \tag{16}$$

where $\mathrm{SD}(x_t^{\circ})$ represents the standard deviation of the predicted variable, terms "$\max(x_t^{\circ})$" and "$\min(x_t^{\circ})$" indicate the maximum and minimum values of the predicted variable. The range/region between the 75th and 25th percentiles of the data distribution is denoted by $Q_3 - Q_1$. Notably, Q_3 and Q_1 can be obtained by applying the inverse cumulative distribution function at cumulative probabilities of 0.75 and 0.25, respectively.

(7) *Coefficient of Determination* (R^2) ($0 \le R^2 \le 1$)
The coefficient of determination, often known as R^2, is a statistical metric that quantifies the extent to which the independent variables can account for the variability in the dependent variable. It evaluates the alignment between predicted and actual values, ranging from 0 to 1. A higher R^2 value indicates a stronger correspondence between the model and the data. The R^2 can be calculated using the following equation:

$$R^2 = 1 - \frac{SS_{\mathrm{res}}}{SS_{\mathrm{tot}}} = 1 - \frac{\sum_{t=1}^{T}\left(x_t^{*} - x_t^{\circ}\right)^2}{\sum_{t=1}^{T}\left(x_t^{*} - \frac{1}{T}\sum_{t=1}^{T} x_t^{\circ}\right)^2} \tag{17}$$

where SS_{res} and SS_{tot} represent sum of squares of residuals and total sum of squares, respectively.

2.3.2. Computational efficiency and convergence

Efficiency and convergence metrics in computational performance aim to evaluate both solution time and solution convergence aspects of OPF learning models. The computational efficiency metrics offer valuable insights into the efficiency and speed of model execution, particularly when compared to computationally expensive approaches. The computational efficiency is typically analyzed via *average computation time*, *speedup factor*, and *time-saving percentage*. Assuming that CT_t° and CT_t^{*} denote the computation time for solving the OPF problem using a learning model and a benchmark solver, respectively, the computational efficiency metrics are described as follows:

$$\textit{Average computation time} = \frac{1}{T}\sum_{t=1}^{T} CT_t^{\circ} \tag{18}$$

$$\textit{Speedup factor} = \frac{\frac{1}{T}\sum_{t=1}^{T} CT_t^{*}}{\frac{1}{T}\sum_{t=1}^{T} CT_t^{\circ}} \tag{19}$$

$$\textit{Time-saving percentage}(\%) = \left(1 - \frac{\frac{1}{T}\sum_{t=1}^{T} CT_t^{\circ}}{\frac{1}{T}\sum_{t=1}^{T} CT_t^{*}}\right) \times 100\ \% \tag{20}$$

Lastly, solution convergence focuses on evaluating the convergence behavior of learning frameworks that employ iterative optimizers, such as algorithms that anticipate active constraint sets, utilize warm-start techniques, or integrate learning-assisted distribution optimization approaches. Assessing solution convergence, primarily measured by the "*number of iterations until convergence*," provides valuable insights into the efficiency and effectiveness of the developed model compared to benchmark solvers, ensuring the attainment of a converged solution. Evaluating solution convergence is essential for comprehending the model's capacity to achieve a stable and accurate solution within an acceptable number of iterations or a specified time frame.

3. End-to-end (E2E) OPF learning schemes

In this section, the current trends and advancements in E2E learning frameworks designed for the OPF problem are presented. As previously discussed, these frameworks can be classified into three separate classes (supervised mapping, unsupervised mapping, and reinforcement learning) based on their learning mechanism and the type of dataset employed during the learning phase. In the context of the E2E OPF learning, the literature predominantly represents the OPF problem as supervised mapping and reinforcement learning tasks, with only a few studies opting for unsupervised mapping algorithms [44–46], which are specifically confined to generative adversarial networks. In the subsequent sections, a breakdown of recent developments in supervised mapping, unsupervised mapping, and reinforcement learning is presented.

3.1. Supervised mapping

The entire E2E works examined in this section are classified under the supervised learning domain, where there is an explicit pairing between input data (e.g., load and RESs) and corresponding OPF results (e.g., generator's output power). The advancement of E2E supervised mapping schemes is introduced under the umbrella of unconstrained, constrained, and graph learning. Notably, in unconstrained, constrained, and graph learning frameworks for E2E mapping, the OPF problem is approached as a regression task aiming to predict continuous variables.

Fig. 2 illustrates the fundamental workflow diagram of the E2E OPF learning process treated as a regression task. Although this basic diagram may accommodate additional steps and complexities, it provides essential insights into the methodology. Initially, a comprehensive set of power system features, encompassing power network topology, generator characteristics, transmission line parameters, and load profiles, is gathered. Subsequently, each sample within the collected dataset undergoes processing through a consensual solver (e.g., IPOPT) to obtain the optimal generation dispatch corresponding to each sample. However, these initial steps can be omitted if a pre-existing dataset of input features paired with generation dispatch results is available. With the input features already paired with generation dispatch results, a machine learning model (e.g., Neural Network (NN)) is trained with a particular objective (e.g., minimizing the L_2-distance between predicted and actual values). The trained machine learning model can then be used to predict the generation dispatch of the OPF problem for a new set of features. This approach bypasses the need for explicit optimization using mathematical programming tools, resulting in a remarkably fast acquisition of OPF solutions for new input features. The following three subsections provide an overview of recent advancements in unconstrained, constrained, and graph OPF learning as a regression task.

3.1.1. Unconstrained learning

Unconstrained E2E learning models learn the mapping from input features to the generation dispatch (predicted variables) by overlooking the OPF technical constraints within the learning phase. In recent years, the deployment of unconstrained E2E in OPF has gained prominence, and a comprehensive overview of these applications is provided in Table 2. This table organizes the OPF type (AC-OPF or DC-OPF), the size of test systems, input features, dataset size, the variation range of input features, learning algorithm, performance evaluation metrics,

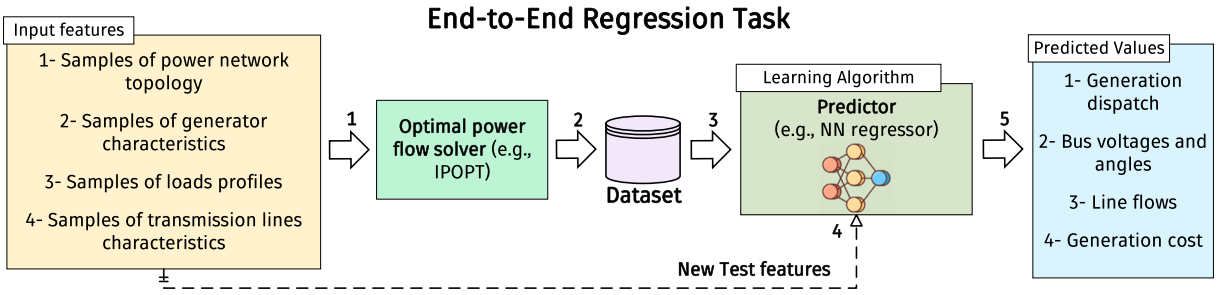


Fig. 2. Basic workflow diagram of the E2E OPF learning as a regression task.

security analysis, and open-source code availability. The size of test systems is quantified by the number of buses, which indicates the scale of the power systems tested. Input feature and its variation range designate which parameters feed the learning algorithm and how much they have been varied to represent the dispersion of the system's states. The learning algorithm indicates the applied machine learning model to perform the regression task. The evaluation metrics involve performance assessment criteria that have been leveraged to demonstrate the benefits of the developed model. The security analysis highlights whether the approach addresses security-constrained OPF, specifies the type of contingencies (line or generator), and mentions the number of contingencies considered (N-1, N-2, etc.).

As one of the early works in this domain, authors in [47] formulated the OPF problem as a regression task using different learning algorithms, including linear and piecewise regression, Gaussian process and NN where the input feature and the predicted variable were load profiles and dispatch cost, respectively. The authors demonstrated the fair potential of NNs in reaching high-accuracy results and immense computational gain of linear regression. Although this work sheds light on the significant capabilities of machine learning regression models in predicting the OPF dispatch cost, it does not delve into retrieving other essential variables inherent in the OPF problem, such as active and reactive power flow at each line. Diverse approaches have thus emerged in recent years to predict and retrieve OPF variables within unconstrained E2E OPF learning, as detailed in the following.

One of the approaches in this realm was to directly predict active and reactive (in the case of AC-OPF) power generation dispatch of generators in OPF test systems by feeding input features (e.g., load profiles) into the learning algorithms, which has been done in [48–51]. By merely focusing on the accurate prediction of power generation dispatch, none of these works approached retrieving other OPF variables which are essential to check the feasibility of generated solutions. It is worth mentioning that in [50], the authors proposed a load encoding scheme for dimension reduction of the OPF learning in large power networks which typically requires massive datasets for accurate learning. The load encoding provides remarkable improvements in both training convergence and prediction accuracy of power generation dispatch, particularly for large-scale test cases.

Within this area, another approach was to predict active power of generators and voltage magnitude at generator buses given input features (load profiles) [9,52–55]. In [52], the authors assessed the predictive performance of different NN models such as fully connected, convolutional and graph NNs (GNNs) to predict active power of generators and voltage magnitude at generator buses for given load profiles and line contingencies under fixed and varying topology. In terms of prediction accuracy, the study's findings indicate that fully connected NNs outperform other NN types when the topology is fixed, while GNNs demonstrate superior performance under varying topology conditions. Without dealing with retrieving the remaining OPF variables [52,53], the authors in [9] approached recovering all OPF variables by feeding predicted variables (active power of generators and voltage magnitude at generator bus) into a power flow solver, given that the predicted variables are feasible operating points. To impose the feasibility of predicted solutions, the authors (*i*) restricted the predicted variables to remain within the permitted bounds using sigmoid activation functions and (*ii*) trained the NN with strictly feasible solutions. Closely related to [9], a similar approach was implemented in [54], [55] to recover full set of OPF variables. In particular, the authors in [55] introduced a meta-learning framework to address the challenge of fully retraining a deep NN when adapting to a new topology by enabling rapid retraining with meta-learning techniques. Meta-learning, often referred to as *learning-to-learn* or *few-shot learning*, is a branch of machine learning that minimizes the training effort required to retrain and adapt algorithms (e.g., deep NNs) for new tasks, such as handling new system topologies. The study in [55] demonstrated that meta-learning *i*) outperforms the training from scratch strategy in terms of predictive accuracy under a new topology with limited data, and *ii*) greatly reduces the time required for adapting NNs to new system configurations.

Another alternative was to employ the machine learning algorithm to learn the mapping between load profiles and bus voltages (magnitude and angles) and, accordingly, retrieve the remaining OPF variables by solving a set of power flow equations [56–62]. In [56], the authors proposed an approach to predict bus voltages of power grids via training a convolutional NN fed by load profiles and network topology. In [57,58], the potential of various learning algorithms, including random forest, multi-target decision tree, and extreme learning machine to predict bus voltages from given load samples, was tested. In addition to predicting bus voltages in [59–61], the authors put forth a post-processing procedure to enhance the feasibility of derived solutions in terms of inequality constraints. The authors' findings demonstrated that the post-processing procedure considerably improved the solution feasibility and could be an essential step in unconstrained E2E approaches where the learning algorithm is unaware of all inherent constraints in the OPF problem.

Instead of relying solely on predicting a limited subset of variables, numerous studies have adopted the approach of high-dimensional OPF prediction, which encompasses all primary variables involved in the OPF problem (active and reactive power of generators, voltage and phase angle of buses) [63–65]. Implementing high-dimensional OPF prediction, which encompasses all primary variables, poses significant challenges in terms of data availability and quality, feature selection, model complexity and scalability, and computational efficiency. These obstacles may result in inaccurate predictions and substantial constraint violations, particularly in large power networks [63]. To deal with such issues, a "compact learning" framework was proposed in [63] to compress the dimension of output space (predicted variables) leveraging "principal component analysis." The authors demonstrated that "compact learning" outperforms current methods in terms of optimality gaps and constraint violations, often by a substantial margin.

3.1.2. Constrained learning

Constrained E2E learning models in OPF account for the technical constraints governing the predicted variables during the learning phase.

Table 2
Application of unconstrained supervised mapping on the OPF problem.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[47]	AC	{5, 24, 73}	Load	[20 %, 200 %]	20 k–185 k	NN, etc.*	1- Optimality, 2- Feasibility, 3- Computation time, 4- Time-saving	–	–
[48]	AC	{14, 73}	Load, line outage	N.A	1 k	NN, random forest	R^2	Line outage (N-1)	–
[49]	AC	{30, 57, 118, 300}	Load	±10 %, ±30 %	200 k	NN	MAPE	–	–
[50]	AC	24–3375 (sixteen test cases)	Load	±20 %	2 k	Linear regression, NN	1- MAE, 2- MSE	–	–
[51]	DC	{14, 118, 300}	Load, RES	10 %	10 k–30 k	NN	1- Optimality, 2- MAE, 3- MAPE, 4- MSE, 5- Computation time	–	–
[52]	AC	24–2853 (eleven test cases)	Load, line outage	±20 %	10 k	NN, GNN	1- Computation time, 2- MSE	Line outage (N-1)	For dataset creation
[53]	DC	{9, 24, 118, 242}	Load	±5 %	200-15 k	NN	1- MAPE, 2- Computation time	–	–
[9]	AC	{39, 57, 118}	Load	±70 %	100 k	NN	1- Optimality, 2- Feasibility, 3- Speedup factor	–	–
[54]	AC	{24, 300, 1354}	Load	[0 %, 140 %]	55 k	NN	1- MAE, 2- MAPE, 3- Relative error, 4- Optimality, 5- Feasibility, 6- Computation time, 7- Speedup factor	–	–
[55]	AC	{14, 30, 118}	Load, line outage	±70 %	100 k	NN	1- MSE, 2- MAPE, 3- Optimality, 4- Computation time, 5- Speed-up factor	Line outage (N)	–
[56]	AC	30	Load, network topology	N.A.	8 k	NN	1- R^2, 2- Computation time, 3- Speedup factor	–	–
[57]	AC	{500, 4918}	Load	±5 %	10 k–40 k	Random forest, etc.*	1- Optimality, 2- Feasibility, 3- Computation time, 4- Speedup factor	–	–
[58]	AC	500	Load	Historical data	~8.7 k	Random Forest	1- MSE, 2- Constraint Violation, 3- Optimality, 4- Computation time	–	–
[59]	AC	{9, 57, 200}	Load, line outage	±20 %	100 k–225 k	NN	1- Optimality, 2- Feasibility, 3- Speedup factor	Line outage (N-1)	Yes
[60]	AC	{118, 300, 2000}	Load	±10 %	40 k–60 k	NN	1- Optimality, 2- Speedup factor, 3- Feasibility	–	Yes
[61]	AC	{39, 300}	Load	N.A.	~52 k–~110 k	NN	1- Optimality, 2- Computation time, 3- Feasibility	–	–
[62]	AC	22883	Load, {line, load, generator} outage	N.A.	~8.5 k	NN	1- RMSE, 2- Feasibility	Line, load, generator outage (N-3)	–
[63]	AC	300–30000 (nine test cases)	Load	±15 %, [80 %, 100 %]	22 k	NN	1- Optimality, 2- Feasibility, 3- Computation time	–	–
[64]	AC	{300, 1354, 1737}	Load	±20 %, [60 %, 100 %]	~5.2 k	NN	1- MAE, 2- Feasibility	–	–
[65]	AC	{118, 300, 1354}	Load	[70 %, 150 %]	21 k	NN	1- MAE, 2- Computation time	–	–

*: Linear and piecewise regression, Gaussian process; ⋆: decision tree, extreme learning machine

In contrast to unconstrained approaches, constrained E2E models ensure adherence to involved constraints, promoting more feasible and reliable solutions. Over the past years, the utilization of constrained E2E methods in OPF has seen significant growth, and an exhaustive compilation of these applications is presented in Tables 3, 4, and 5. In addition to unconstrained E2E methodologies, where the direct prediction of variables and subsequent post-processing steps held significance, the realm of constrained learning frameworks stands out by virtue of the diverse techniques in which the desired constraints are assimilated into the learning process. These techniques encompass various methods such as the penalty method, the Lagrangian duality approach, the innovative use of implicit differentiable layers, and more, as expounded below.

The frequently utilized penalty method incorporates an additional term with penalty coefficients into the main loss function of the learning algorithm, serving to penalize any violation of constraints and promoting the regression model to output solutions that adhere to the penalized constraints. The penalty method has been leveraged in numerous studies to encourage the learning algorithm to observe the OPF constraints, each emphasizing particular limitations [66–75]. Table 3 provides particular attributes of these works. In [66–68], the authors designed the loss function of the learning algorithm as the weighted sum of *i*) the MSE of the predicted variables (as in unconstrained E2E) and *ii*) the penalty term associated with the violation of transmission line limits in a DC-OPF. In [69], the authors augmented the loss function formulated in [66–68] with an extra penalty term taking into account the violation of the boundaries of the slack bus generation. Accordingly, various penalty terms linked to other OPF constraints, including voltage and active/reactive power generation boundaries [70–72] as well

Table 3
Application of constrained supervised mapping on the OPF problem leveraging penalty method.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[66]	DC	{30, 118, 200, 300}	Load	[100 %, 130 %]	30 k	NN	1- Dispatch Cost, 2- Feasibility, 3- Computation time, 4- Speedup factor	–	–
[67]	DC	{30, 57, 118, 300}	Load	±10 %- check [59] for further variations	55 k	NN	1- Optimality, 2- Dispatch cost, 3- Feasibility, 4- Computation time, 5- Speedup factor	–	–
[68]	DC	{30, 57, 118, 300}	Load	±10 %	20 k–60 k	NN	1- Optimality, 2- Feasibility, 3- Computation time, 4- Speedup factor	–	–
[69]	DC	{30, 118, 300}	Load	[100 %, 130 %]	60 k	NN	1- Feasibility, 2- Optimality, 3- MAE, 4- Speedup factor, 5- Computation time	–	-
[70]	AC	{14, 30, 57, 118, 200}	Load	±30 %	200 k	NN	1- Optimality, 2- Feasibility, 3- Computation time	–	–
[71]	AC	{30, 118}	Load	±10 %	~81 k–95 k	NN	1- Optimality, 2- Feasibility	–	–
[72]	AC	{12, 14, 30, 57, 118}	Load	±20 %	~1 k–~2 k	NN	1- Optimality, 2- Feasibility, 3- Computation time	–	–
[73]	DC	{39, 118, 162}	Load	[60 %, 100 %]	100 k	NN	1- MAE, 2- Feasibility, 3- Optimality	–	Yes
[74]	AC	{14, 39, 118, 162}	Load	[60 %, 100 %]	10 k	NN	1- MAE, 2- Feasibility, 3- Optimality	–	Yes
[75]	AC, DC	{39, 57, 118, 162}	Load	[60 %, 100 %]	10 k	NN	1- MAE, 2- Feasibility	–	Yes

Table 4
Application of constrained supervised mapping on the OPF problem leveraging Lagrangian duality method.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[77]	AC	{30, 118}	Load	±10 %	5 k	NN	1- Dispatch Cost, 2- Feasibility, 3- Computation time	–	–
[78]	AC	{14, 30, 39, 57, 73, 89, 118, 162, 189, 300}	Load	±20 %	~69 k–~395 k	NN	1- MAPE, 2- Feasibility, 3- Computation time	–	–
[79]	AC	{30, 118, 300}	Load	±20 %	4 k	NN	1- MAE, 2- Feasibility	–	–
[80]	AC	{89, 118, 300, 403, 1737, 3411}	Load	[[80 %, 90 %], [110 %, 120 %]]	2 k	NN	1- MAE, 2- Feasibility, 3- Optimality, - Computation time	–	–
[81]	AC	14–3375 (twenty-four test cases)	Load	±20 %	2 k	NN	1- Optimality 2- Computation time 3-MSE	–	–
[82]	AC	{1737, 3411, 6705}	Load	Check [74] for details	10 k	NN	1- MAE, 2- Optimality, 3- Feasibility, 4-Computation time	–	–
[83]	AC	{118, 162, 300}	Load	±20 %	N.A.	NN	1- MAE, 2- Optimality	–	–
[84]	DC	{14, 57}	Load	Check [76] for details	4 k	NN	1- RMSE, 2- Feasibility, 3- Computation time	–	–
[85]	AC	{89, 118, 300}	Load	[80 %, 100 %]	N.A.	NN	1- Optimality 2- Feasibility	–	–
[86]	AC	{57, 118}	Load	±30 %	30 k	NN	1- Optimality 2- Dispatch Cost, 3- Feasibility	–	–
[87]	AC	39	Load	N.A.	1 k	NN	1- Optimality 2- Feasibility, 3- Computation time, 4- MSE	–	–

as Karush–Kuhn–Tucker (KKT) [73–75] conditions, have been incorporated into the training loss function to diminish constraint violations. While the results of the above studies demonstrated substantial improvement in the feasibility of generated solutions compared to unconstrained E2E, the feasibility is still not guaranteed. To improve the feasibility, a post-processing procedure was proposed in [67]. Furthermore, to ascertain the worst constraint violations, the method outlined in [76] was utilized to reformulate the trained NN as a mixed-integer linear programming problem in [73], [74]. More interestingly, while works [73,74] determined the worst constraint violations for a trained NN, the

Table 5
Application of constrained supervised mapping on the OPF problem leveraging implicit layers.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[88]	DC	{39, 118, 162}	Load	[60 %, 100 %]	42 k	NN	1- Optimality 2- Feasibility 3- MAE 4- RMSE 5- Speedup factor	–	–
[39]	AC	57	Load	[30 %, 170 %]	1.2 k	NN	1- Optimality 2- Feasibility 3- Computation time	–	Yes
[89]	DC	200	Load	±10 %	200	NN	1- Optimality 2- Feasibility 3- Computation time	–	–
[90]	DC	118	Load	±5 %, ±10 %	50 k	NN	1- Optimality 2- Computation time	–	Yes

framework to minimize worst-case violations with the training process of an NN was embodied in [75].

Another well-documented method to integrate operational constraints into OPF learning tasks is the Lagrangian duality method, as extensively discussed in prior research [77–87]. The key attributes of the works in question are detailed in Table 4. This method involves the incorporation of Lagrange multipliers related to certain or all problem constraints into the primary loss function as penalty terms, which encourages the learning algorithm to output solutions that minimize constraint violations while generating high-quality predictions. In [77], the authors combined deep NN learning with the Lagrangian duality method to capture inequality constraints in the OPF problem, whereas a broad body of literature leveraged the same approach to tackle all equality and inequality constraints [78–86]. A significant hurdle encountered in penalty and Lagrangian duality approaches revolves around the selection of optimal penalty coefficients and Lagrangian multipliers. This selection aims to strike the optimal balance between prediction accuracy and constraint violation. Though past studies utilizing the penalty method have empirically tackled this challenge [68], the proposal outlined in [78] and [86] take distinct paths. The approach in [78], by integrating a subgradient method, dynamically adjusts Lagrangian multipliers throughout the training process, yielding more robust solutions. Alternatively, the approach in [86] leverages primal and dual NNs to predict Lagrangian multipliers. Apart from the aforementioned studies focused on lowering the infeasibility rate, a recent investigation challenged the design of the NN architecture by analyzing the type of loss function and the number of neurons employed, utilizing the Lagrangian duality method [87]. The authors utilized two loss functions: the well-known MSE between predicted and actual dispatch decisions and a novel “decision loss” aimed explicitly at minimizing dispatch costs. They also examined how the number of neurons in the NN architecture affects the optimality gap and constraint violation rate. Experiments conducted on a 39-bus system revealed that the decision loss outperforms MSE in both optimality and feasibility. Finally, the authors concluded that increasing the number of neurons can potentially reduce the constraint violation rate, although no definitive relationship between the number of neurons and the optimality gap could be established.

Recently, substantial interest has been directed toward structured NN layers, widely known as implicit layers [39,88–90]. These layers are designed to delineate implicit interconnections between input and output elements. Generally, there exist two types of NN layers: explicit and implicit. Explicit layers are characterized by a particular computation (explicit functions like linear, convolutional, and ReLU functions) that takes the input data and produces the corresponding output. In contrast, implicit layers are established based on a collection of conditions (constraints) that the output is required to fulfill [91–94]. Hence, capitalizing on implicit layers lays the foundation for E2E learning that inherently fulfills problem constraints. In [88], the authors added a differentiable projection layer at the last layer of a deep NN, acting as an implicit layer, to map hidden vectors into the feasible region for E2E DC-OPF learning. Despite achieving complete feasibility in their empirical investigations, the procedure’s computational intensity is tremendously heightened by the presence of an optimization layer. In [39], the authors incorporated an implicit layer called “equality completion” to guarantee the satisfaction of equality constraints in E2E AC-OPF learning. In this procedure, an NN initially generates a subset of variables, followed by the subsequent inference of the remaining variables through explicit determination or the resolution of implicit equations. Later on, a differentiable corrective procedure rooted in the principles of gradient descent was formulated to handle situations involving violations of inequality constraints, although without any guarantees. Closely related to [39], an analogous procedure known as “constraint completion” was developed in [89] to manage equality constraints, combined with the integration of a gauge mapping module to ensure the fulfillment of inequality constraints within the context of DC-OPF. Similarly, the gauge mapping module as an implicit layer was leveraged to ensure the feasibility of solutions in a two-stage DC-OPF in [90].

3.1.3. Graph learning

In general, graph learning pertains to applying machine learning techniques to graph-based data, where graphs represent interconnected data formed by a network structure like traffic networks and power grids [95]. Graph learning is the practice of utilizing graph representations to incorporate the power system’s topology into the learning process, allowing for effective management of dynamic grid topology changes, notably during contingencies. While the previously discussed learning algorithms show promise in predicting the OPF solution in normal operation conditions, they require retraining with new datasets to adapt to changing grid topologies during contingencies. This deficiency in adaptability to grid topology significantly hinders their acceptance among grid operators, primarily due to computational constraints. GNN, as a specific instance within the realm of NNs, has been widely leveraged as a suitable candidate to integrate grid topology into the OPF learning process to ensure robustness and adaptability in the face of grid topology alterations. Without loss of generality, GNNs have undergone extensive development and refinement for OPF learning with a primary emphasis on respecting the grid topology [45], [52], [96–100]. Various attributes of these works are listed in Table 6. In [45] and [52], the authors utilized GNN for learning OPF, where the GNN is not topology-dependent due to the utilization of architectural components tied to the global structure, making them particularly effective only for single grid topology scenarios. In [96], a robust graph learning model was engineered by exploiting the locality property and implementing custom graph-based layers. This model showcased resilience when confronted with varying grid topologies. Utilizing the locality property of locational marginal prices and voltage magnitude, an adaptive graph learning model for accommodating grid topology changes in AC-OPF was proposed in [97,98]. A physics-guided GNN for addressing OPF by incorporating physics-embedded graph convolution kernels and iterative feature construction techniques was introduced by [99]. Finally, a physics-guided GNN combined with Lagrangian duality to incorporate OPF constraints into the learning process was suggested in [100]. The

Table 6
Application of graph supervised mapping on the OPF problem.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[45]	AC	{30, 118}	Load	±10 %	101 k	GNN	1- Feasibility, 2- Dispatch cost	–	–
[52]	AC	{24, 30, 39, 57, 73, 118, 162, 300, 588, 1354, 2853}	Load, line outage	±20 %	10 k	NN, GNN	1- Computation time, 2- MSE	Line outage (N-1)	Dataset generation
[96]	AC	{24, 30, 39, 57, 73, 118, 162, 300, 588, 1354, 2853}	Load, etc.*	Check [88] for details	15 k	GNN	NRMSE	Line outage (5-20 contingencies)	–
[97]	AC	{118, 1354, 2383}	Load, Generator cost coefficients	±10–30 %	10 k	GNN	1- MSE, 2- Feasibility, 3- Computation time	Line outage (1-4 contingencies)	Yes
[98]	AC, DC	{118, 2383}	Load, Generator cost coefficients	N.A.	N.A.	GNN	1- MSE, 2- Feasibility	Line outage (N-1, N-2)	–
[99]	AC	{39, 57, 118, 300}	Load, RES	Load = ±10 %, Check [91] for RES	12 k	GNN	MAE	Line outage (N-1)	–
[100]	AC	{30, 118}	Load, RES	Load = [60 %, 100 %], RES = [0 %, 100 %]	100 k	GNN	1- Feasibility, 2- Computation time, 3- Dispatch cost, 4- Renewable curtailment	Line outage (N-1)	–

*: Line elements, generators active power, generators voltage magnitude, transformers off-nominal turns ratio.

results in the above studies demonstrated high capability of graph learning models in dealing with changes in grid topology while maintaining high predictive ability.

3.2. Unsupervised mapping

Unsupervised machine learning models are applied in scenarios where a direct correspondence between input features and OPF solutions is not explicitly defined. While the majority of unsupervised models are typically utilized for clustering, dimensionality reduction, and association tasks, which are not directly aligned with E2E OPF mapping, generative adversarial networks [36] as the predominant unsupervised learning model can serve for OPF mapping. The generative adversarial network is structured with two main modules: the *generator* and the *discriminator*, each performing distinct functions within the network for unsupervised OPF mapping. The generator's primary objective is to acquire knowledge about the distribution of OPF decisions through adversarial interactions with the discriminator. During the training process, the generator generates a synthetic solution based on given input data. Conversely, the discriminator aims to classify whether the generated solution by the generator is synthetic or actual. In this dynamic interaction, the generator strives to minimize the detection of synthesized solutions by the discriminator, while the discriminator aims to accurately distinguish between synthesized and real solutions. This adversarial interplay facilitates the generator's understanding of the OPF solution distribution, enabling it to generate solutions for a new set of input data. However, due to the lack of control over the generated solution in adversarial learning, researchers have developed conditional generative adversarial networks to condition generated samples that serve as a suitable fit for OPF mapping when operational constraints need to be satisfied. The utilization of conditional generative adversarial networks in the context of OPF mapping has been examined in [44–46]. In [44] and [45], conditional generative networks were designed for AC-OPF and DC-OPF mapping, respectively, where feasibility layers were added to the structure of the generative model to ensure the feasibility of generated solutions. A conditional generative model for AC-OPF mapping, proposed in [46], incorporated power flow constraints by penalizing constraint violations during the training process by introducing penalty terms. This approach ensures the feasibility of solutions while maintaining a low optimality loss. Despite sporadic instances highlighting the applicability of generative adversarial networks for OPF mapping, there is a necessity for in-depth exploration of various aspects of generative models in the context of OPF mapping.

3.3. Reinforcement learning

Reinforcement learning is a promising machine learning tool that has lately attracted immense attention in different fields due to its auspicious decision-making abilities [25]. It offers the benefit of autonomous learning through iterative trial-and-error within a dynamic system (environment), making it a valuable tool for addressing a broad spectrum of practical control decision challenges under uncertainty [22]. The basic process of learning about the dynamic interaction between an *agent* and its operating *environment* is at the core of reinforcement learning. Through interaction with the environment and the receipt of feedback, the agent investigates, decides, and fine-tunes its control strategy in pursuit of a specific goal, namely, maximizing a user-defined reward. In a standard reinforcement learning procedure, the agent's course of action is determined by its policy and the current state, followed by the environment transitioning to a new state. This strong self-learning ability of reinforcement learning has been widely tested and analyzed on the OPF problem. By formulating the OPF problem as a Markov decision process, the operator of the power system assumes the role of the agent, while the power flow solver operates as the environment. During the training process, the agent perceives the system's state, executes an action (e.g., setting generator active power levels), obtains a reward (e.g., reflecting dispatch costs), and subsequently refines its policy based on the acquired reward signal. This iterative cycle enables the agent to learn and improve its decision-making capabilities toward maximizing the reward. Accordingly, once this model is trained,

the agent's decision-making becomes notably rapid, positioning it as a compelling prospect for achieving fast OPF solutions.

While classical reinforcement learning methods demonstrate strong performance in various domains, they face significant challenges when dealing with expansive state and action spaces. This is especially evident in scenarios involving mixed continuous and discrete actions, such as the OPF problem, where the complexity of high-dimensional data is formidable. As a response to this challenge, researchers have turned their attention to deep reinforcement learning, a paradigm that capitalizes on advanced function approximation techniques. Deep reinforcement learning achieves this through deep learning models, like NNs, enabling the efficient estimation of value functions and policies. OpenAI's open-source deep reinforcement learning platform (Gym) [101] has played a pivotal role in the widespread application and development of diverse deep reinforcement learning algorithms. These include Deep Deterministic Policy Gradient (DDPG) [102–105], Twin Delayed Deep Deterministic Policy Gradient (TD3) [105–108], Soft Actor-Critic [109–113], and Proximal Policy Optimization (PPO) [114–117], all of which have been effectively applied to tackle the complexities of the OPF problem.[4] Despite several studies comparing the performance of the above-mentioned algorithms [108–110,112], there remains no definitive consensus on which of these algorithms is the most effective, as per the authors' current understanding. In the context of evaluating the effectiveness of these algorithms, it is noteworthy to introduce Grid2Op [118] as a pivotal open-source platform that provides a carefully built simulation environment for creating and thoroughly testing various types of decisions and policies, e.g., reinforcement learning algorithms, for power grid management.

As previously elucidated, the primary mission of a reinforcement learning agent, embodying the role of a power system operator, resides in acquiring and enhancing its decision-making prowess. This is achieved by internalizing a policy that strategically guides its actions to maximize the accrued reward extracted from the environment. While conventional deep reinforcement learning algorithms generally allow agents to freely explore control actions, this approach may lead to violations of operational constraints during online dispatch. The design of the environment in reinforcement learning modules is crucial for ensuring high-quality decision-making. A recent study [102] analyzed the impact of various environmental parameters, such as training data, reward function choice, episode structure, and observation space, on OPF-learning tasks. The study presented three key insights: *i)* using realistic time-series data is strongly recommended over artificially generated data, *ii)* expanding the observation space with redundant information, like voltage values, does not appear to provide significant benefits, and *iii)* the choice of reward function should be tailored to the specific problem.

Similar to the context of constrained E2E learning, various methods are employed to direct the agent in making well-informed decisions concerning the OPF operational constraints. These approaches encompass techniques such as the penalty method [103,105–107,109,111,114–117], Lagrangian duality [104,108], and implicit layers like the convex safety layer [110] and the holomorphic embedding-based layer [112]. The penalty method integrates additional components into the primary reward function where newly introduced elements are intricately crafted to address and impose penalties on any instances of power system constraint violations [103,105–107,109,111,114–117]. In the Lagrangian duality method, the Lagrangian multipliers associated with system constraints are cleverly added to the reward function [104,108]. Moreover, a graph reinforcement learning approach integrated with meta-learning for multi-stage OPF was proposed in [113] to improve the generalizability of reinforcement learning models when facing unexpected changes in grid topology. The study demonstrated how meta-learning facilitates the quick adaptation of reinforcement learning models to unforeseen grid topology reconfigurations. Demonstration of the implementation reveals that a proficiently trained reinforcement learning agent can swiftly generate solutions, maintaining a minimal optimality gap and ensuring no constraint violations [108,110]. Table 7 lists different attributes of works employing reinforcement learning for the OPF.

[4] Specific classifications and intricate details of various deep reinforcement learning algorithms have been extensively addressed in the existing literature and are beyond the scope of the present work. Readers seeking in-depth information can refer to [22] for further insights.

4. Learning-to-optimize (L2O) OPF schemes

This section elucidates the current trajectory of advancements in L2O frameworks specifically designed to tackle the OPF problem. The goal of these frameworks is to supplement conventional OPF optimization models with learning algorithms. As previously discussed, these frameworks can be systematically classified into three categories: learning active set of constraints, warm-start, and learning-assisted distributed/simplified optimization. An overview of these frameworks is presented in the following three subsections.

4.1. Learning active set of constraints

The concept of "*Learning active set of constraints*" revolves around removing non-binding (inactive) constraints in the OPF problem that inherently do not affect the optimal operating point. The OPF problem is cast as an optimization challenge governed by both equality (1b)–(1e) and inequality constraints (1f)–(1j). In the context of an optimal OPF solution, all equality constraints are active, meaning they are binding and fully satisfied at the optimal point. However, only a subset of inequality constraints becomes active at this point, signifying that their boundaries encompass the optimal solution [40]. Consequently, if an inequality constraint remains inactive at the optimal solution, it implies that including this constraint in the formulation was unnecessary and could be deemed redundant. In the field, there is a widespread consensus that a substantial reduction in these inequality constraints can lead to a notable reduction in the OPF computational complexity [119]. This, in turn, yields a more streamlined optimization problem, ultimately resulting in cost-effective solutions.

Fig. 3 presents a basic workflow diagram of learning active set of constraints to downsize the OPF problem. While this fundamental diagram is adaptable to incorporate supplementary steps, it serves as a primary tool for gaining insights into the methodology. To start, a comprehensive array of input features is assembled. These features are then supplied to a consensus solver (e.g., IPOPT) to determine the active set of inequality constraints corresponding to the input features. Once the input features are matched with an assortment of active constraints, a learning classifier like an NN is employed. The primary objective here could involve minimizing the binary cross-entropy loss function, for instance. Once the classifier has been trained, it can be employed to predict the active status of inequality constraints for new test features. Following this, a downsized OPF optimization problem is constructed by preserving the original objective function while incorporating the equality and predicted set of active inequality constraints. A consensus solver is then proceeded to solve the optimization problem. To ensure the achievement of an optimal solution for the entire problem, it is imperative to conduct an iterative feasibility assessment [120]. This is essential because some of active inequality constraints may not be included in the initially predicted set.

Table 8 and 9 present a comprehensive inventory of distinct attributes observed in studies on learning active set of constraints for the OPF problem. In [52,120–123], the authors leveraged different NN classifiers to perform a mapping between input features (e.g., load) and the active set of OPF constraints. For instance, a pair of hierarchical GNNs were utilized in [123] to first predict the congested lines, before feeding this prediction as an input feature for another GNN to predict

Table 7
Application of reinforcement learning on the OPF problem.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Number of Episodes	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[102]	AC	{59, 372}	Load	±30 %	Tuning parameter	DDPG	1-Optimality, 2-Feasibility, 3-Reward of algorithm	–	Yes
[103]	AC	9	Load	N.A.	10 k	DDPG	–	–	–
[104]	AC	118	Load	$\pm\sqrt{5}$%	N.A.	DDPG	1- Dispatch cost, 2- MAE, 3- Time-saving factor, 4- Feasibility	–	–
[105]	AC	{39, 710}	Load	[70 %, 120 %]	10 k–50 k	DDPG	1- Computation time, 2- Rewards of algorithms, 3- Feasibility	Line outage (N-1)	–
[106]	AC	39	Load	N.A.	60 k	TD3	1. Dispatch cost, 2. Computation time	–	–
[107]	AC	118	Load	±20 %	~1.9 k	TD3	1- Dispatch cost, 2- MAE	–	–
[108]	AC	{14, 30}	Load, RES	Historical data	1.5 k	TD3	1- Optimality, 2- Feasibility	–	–
[109]	AC	118	Load	N.A.	20 k	SAC	Reward of algorithms	–	–
[110]	AC	{14, 30, 118, 300}	Load, RES	[50 %, 120 %]	2.4 k	SAC	1- Optimality, 2- Feasibility, 3- Speedup factor	–	–
[111]	AC	{39, 57}	Load	[70 %, 120 %]	5 k	SAC	1- Feasibility, 2- Control Cost, 3- Rescheduling time	Line outage (N cont.)	–
[112]	AC	{30, 118, 300, 1354, 2000}	Load, RES	[50 %, 120 %], [60 %, 100 %]	28.8 k	SAC	1- Optimality, 2- Feasibility, 3- Speedup Factor	–	–
[113]	AC	{39, 126}	Load, RES	Historical data	N.A.	SAC	1- Optimality, 2- Computation time, 3- Reward of algorithms	Line outage (N-1)	-
[114]	AC	{14, 200}	Load	±40 %	~74 k–85 k	PPO	1- Optimality, 2- Feasibility, 3- Reward of algorithms	Line outage (N-1)	–
[115]	AC	200	Load, Wind	Load = ±40 %, Wind = [Min, Max]	165 k	PPO	1- Optimality, 2- Feasibility, 3- Computation time	Line outage (N-1)	–
[116]	AC	30	Load	Check [88] for details	500	PPO	Optimality	Line outage (N-1)	–
[117]	AC	118	Load	N.A.	65 k	PPO	Reward of algorithm	Line outage (N-1)	–

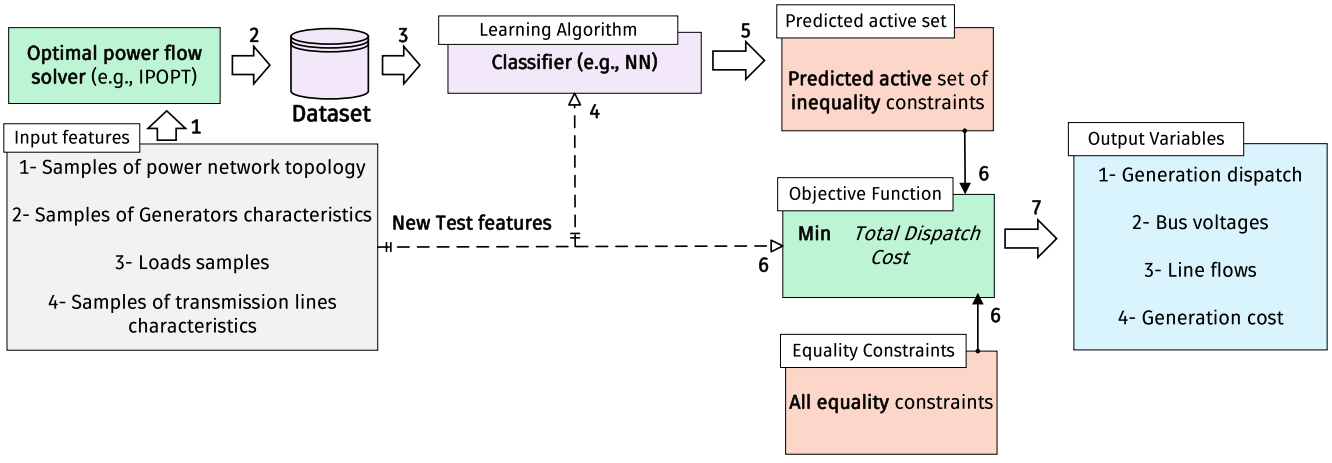


Fig. 3. Basic workflow diagram of learning active set of constraints.

generators running at their maximum limits. This way, non-binding constraints linked to generators' maximum power and transmission lines were filtered out. In a related study, multiple learning classifiers, including linear discriminant analysis, NN, support vector machine, and gradient-boosted decision tree, were tested in [124] to track active inequality constraints of the DC-OPF. The numerical results suggest that linear discriminant analysis not only achieves comparable testing performance in terms of accuracy and run-time to meticulously trained NNs but also outperforms other intricate algorithms in training efficiency. Considering that a security-constrained OPF substantially

Table 8
Application of learning active set of constraints on the OPF problem– Part 1.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[52]	AC	{24, 30, 39, 57, 73, 118, 162, 300, 588, 1354, 2853}	Load, line outage	±20 %	10 k	NN, GNN	Binary cross entropy statistics	Line outage (N-1)	Dataset generation
[120]	AC, DC	{24, 30, 39, 57, 73, 118, 162, 300, 588, 1354, 2853, 4661, 9241}	Load, Maximum Power of generators, Line elements	{±10 %, ±15 %}	1 k–10 k	NN	1- Number of active constraints, 2- Time-saving, 3- Accuracy of active constraint sets prediction	-	Yes
[121]	DC	{24, 30, 39, 57, 118, 162, 300, 1888}	Load	±3 %	50 k	NN	Accuracy of active constraint sets prediction	-	–
[122]	DC	{14, 18, 300}	Load, RES	±10 %	10 k–30 k	NN	1- Accuracy of active constraint sets prediction, 2- Computation time	-	–
[123]	DC	{73}	Load	±10 %	10 k	GNN	1- Accuracy of active constraint sets prediction, 2- Computation time, 3- Optimality	-	–
[124]	DC	162	Load, RES	±6 %	55 k	NN, etc.*	Accuracy of active constraint sets prediction	-	Yes
[125]	AC	{30, 118, 300}	Load	±10 %	20 k	NN	1- Computation time, 2- Solution Convergence, 3- Number of active constraints 4- MSE	Line outage (N-1)	–
[126]	DC	{30, 118}	Load	±10 %	20 k	NN	1- Computation time, 2- Solution Convergence, 3- Number of active constraints	Line outage (N-1)	–

*: Linear discriminant analysis, NN, support vector machine, gradient boosted decision tree.

Table 9
Application of learning active set of constraints on the OPF problem–Part 2.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[127]	AC	{39, 118, 500, 1354}	Load	±30 %, check for further variations	4.2 k–7 k	NN	1- RMSE, 2- Optimality, 3- Solution convergence, 4- Accuracy of active constraint sets prediction, 5- Computation time, 6- Time-saving factor	-	Yes
[128]	DC	{14, 39}	Load	{±20 %, ±50 %, ±80 %}	60 k	NN	1- Feasibility, 2- Computation time, 3- Optimality, 4- MAPE	-	Yes
[129]	AC	{118, 793, 1354, 2312, 4601, 10000}	Load, line limits	Check [130] for details	5 k	NN, etc.*	1- Accuracy of active constraint sets prediction, 2- Computation time, 3- Time-saving	-	–
[130]	DC	1814	Load, RES, system topology	Check [130] for details	1 k	NN	1- Accuracy of active constraint sets prediction, 2- MAE	–	–
[131]	DC	{14, 39, 118}	Load	{±25 %, ±50 %, ±75 %, ±100 %}	10 k	NN, Nearest neighbors regression	1- Accuracy of active constraint sets prediction, 2- Computation time, 3- Optimality	-	Yes
[132]	DC	73	Load	N.A.	~8.7 k	Nearest neighbors regression	1- Accuracy of active constraint sets prediction, 2- Computation time	-	Yes
[133]	DC	{73, 2000}	Load, RES	N.A.	~8.7 k	Nearest neighbors regression	1- Accuracy of active constraint sets prediction, 2- Computation time, 3- Optimality, 4- Infeasibility	-	-

*: Ridge classifier, XGBoost, input convex NN, monotone gradient NN.

amplifies the number of inequality constraints, prior works in [125] and [126] effectively employed NNs to identify active constraints while simultaneously conducting an iterative feasibility analysis. In [127], an NN regressor was employed initially to map load profiles into active and reactive power values for generators, and then, both the load profiles and generator dispatch data were used as inputs for two NN

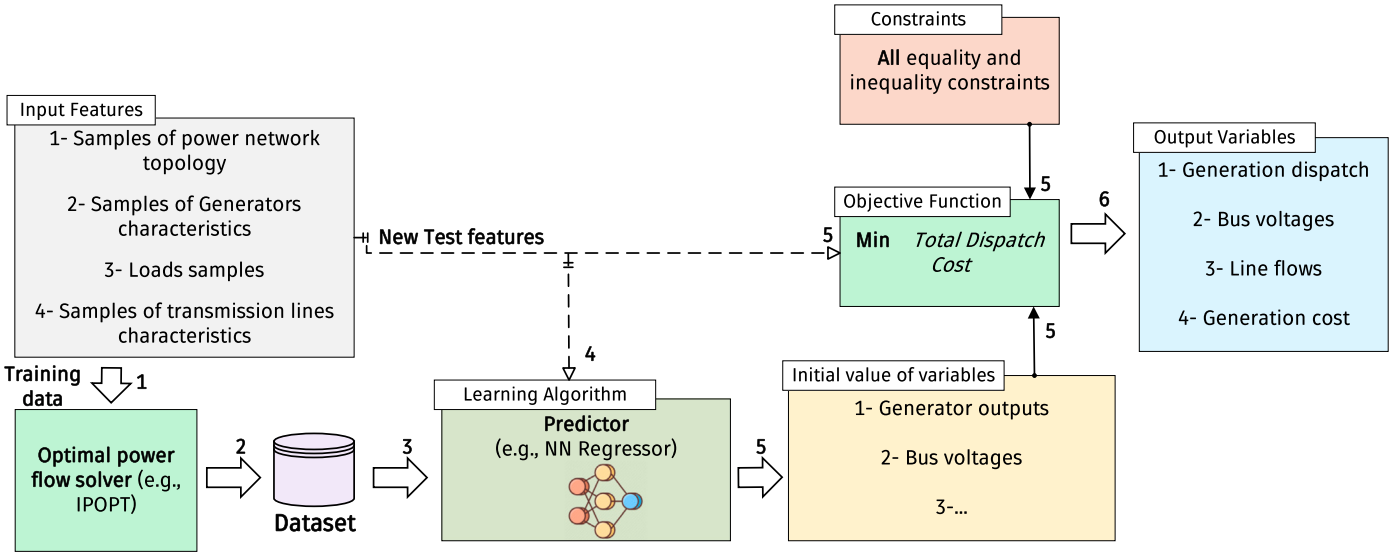


Fig. 4. Basic workflow diagram of the warm-start using learning algorithms.

classifiers to systematically identify inactive voltage and line flow constraints. Differently, an NN was developed in [128] to enable a mapping between the system's load and its dispatch cost. Subsequently, the cost gradient with respect to the load was calculated using the NN. These gradients were then identified as dual variables or locational marginal prices, which were subsequently used to predict the active constraints. In a related approach, [129] proposed an innovative NN architecture, referred to as the mixture of gradient experts, to forecast OPF dual variables, later utilized to screen non-binding constraints. In [130], a physics-aware machine learning model was proposed for the DC-OPF market clearing problem. First, an NN was trained to map the net load and the system topology into the DC-OPF active constraints. These active constraints facilitate simplifying the initial market-clearing optimization into a set of linear equations, enabling the efficient and fast determination of both dispatch decisions and locational marginal prices. Finally, a range of studies focused on applying machine learning algorithms to determine the active constraints in DC-OPF unit commitment problem [131–133]. For instance, [131] trained a deep NN to estimate the system dispatch cost, which was subsequently utilized as an upper bound for the model-based constraint screening problem. Other works [132,133] proposed techniques based on nearest neighbors regression for filtering inactive constraints in a similar problem. In summary, multiple studies have shown that a proficiently trained classifier, capable of predicting all active constraints in OPF, can yield optimal and feasible solutions while significantly reducing the computation time.

4.2. Warm-start

The OPF embodies an intricate nonlinear optimization problem, often characterized by the presence of multiple local optima. It is widely recognized that enhancing convergence and solution quality can be achieved by initializing the nonlinear optimization solver with a well-informed initial guess for decision variables. Warm-starting, characterized by initial values for decision variables, not only expedites convergence but also conserves computational resources. Furthermore, it equips the optimization solver with the capability to navigate away from local optima, ultimately reaching either the global or superior local optimum. The fundamental concept of warm-start, as one of the subclasses of L2O, involves generating an initial estimate using E2E learning models and subsequently initializing the nonlinear optimization solver with these derived values, even though they may not inherently represent feasible solutions. Fig. 4 provides a visual representation of the workflow for implementing warm-starting an OPF solver through machine learning techniques. This process comprises two distinct phases. In the initial phase, similar to E2E learning models, a machine learning model undergoes training on a labeled dataset. This machine learning model acquires the capability to make predictions for target variables, which serves as an informed initial estimate for the optimization solver. Subsequently, leveraging these predicted initial values, the original OPF problem is steered towards expedited convergence, a key advantage of incorporating learning paradigms within the OPF optimization process.

Table 10 presents a compilation of characteristics identified in studies utilizing machine learning algorithms to warm-start the OPF [134–138]. In [134], the authors conducted a runtime comparison between warm-starting the AC-OPF with a GNN and using both a conventional optimization solver and a warm-starting approach based on a DC-OPF solution. The findings highlight the substantial acceleration achieved in AC-OPF computations when employing the GNN-based warm-starting approach in contrast to the established benchmarks. Likewise, in another study [135], the execution time of a warm-start obtained through a machine learning approach, specifically random forest, was compared to warm-starts initiated by DC-OPF and a flat start. The author demonstrated accelerated convergence of the learned warm-start compared to employing either a DC-based warm-start or a flat start, and these advantages seem to vary depending on the solver being utilized. In a different study [136], it was demonstrated that several learning algorithms, as outlined in Table 10, outperformed DC-OPF when it came to initializing AC-OPF with warm-starts. In [137] and [138], the authors employed NNs to predict both the OPF problem and its dual variables. This predictive approach was utilized to initiate the AC-OPF solver, ultimately achieving globally optimal solutions and substantial improvements in computational speed. To summarize, warm-starting can offer a promising strategy for enhancing the efficiency of AC-OPF calculations, provided that the initial guess is sufficiently qualified to effectively steer the optimization solver.

4.3. Learning-assisted distributed/Simplified optimization

The final category of L2O approaches involves harnessing machine learning techniques to enhance distributed and simplified optimization frameworks tailored for solving the OPF problem. In this context, the term "distributed" denotes managing various regions within the power

Table 10
Application of learning algorithms to warm-start the OPF problem.

Ref.	OPF type	No. of buses in test systems	Input feature	Variation of input feature	Dataset size	Learning algorithm	Evaluation metrics	Security analysis	Open-source code
[134]	AC	{200, 2000}	Load	N.A.	N.A.	GNN	Computation time	–	–
[135]	AC	{14, 57, 118, 300}	Load	±20 %	2 k	Random forest	1- Relative Error, 2- Computation time, 3- Solution convergence	–	–
[136]	AC	3	Load	N.A.	600	NN, etc.*	1- Dispatch cost, 2- Feasibility	–	–
[137]	AC	{14, 30, 57, 118, 300}	Load	±10 %	10 k	NN	1- Optimality 2- Feasibility 3- Speedup factor	–	–
[138]	AC	{22, 39, 118}	Load	N.A.	200, 1.9 k, 4 k	NN	1- Dispatch cost, 2- Computation time	–	–

*: Linear regression, support vector machine, nearest neighbors regression, decision tree.

network, which ensures coordinated operation. Conversely, the term "simplified" signifies methodologies that capitalize on relaxation and approximation techniques to effectively address the inherently non-convex and NP-hard characteristics of the OPF problem. A promising trajectory in the evolution of future smart grid systems involves harnessing distributed and decentralized optimization algorithms alongside secure communication protocols to effectively coordinate a diverse array of infrastructural components. Notably, this demands system operators skillfully navigate distributed AC-OPF challenges, particularly when power networks are divided into discrete regions. Achieving this objective encompasses applying various distributed optimization techniques, among which ADMM emerges as a prominent choice. However, it is essential to note that ADMM for distributed OPF is frequently challenged by numerical stability and convergence issues. This is particularly true because AC-OPF is nonlinear and nonconvex, leading to situations where the convergence and optimality guarantees of ADMM may not hold consistently. Several studies have employed learning algorithms to enhance the performance of ADMM-based distributed OPF [15,139,140]. In [139], the authors thoroughly investigated the application of reinforcement learning to address the task of mastering the selection of ADMM penalty parameters. Their findings demonstrated the superior performance of the learning-based approach in comparison to the state-of-the-art method. A distributed DC-OPF algorithm was presented in [140], where ADMM is integrated with a trained recurrent NN to include predictions of dual and consensus variables following a predetermined number of iterations. The authors revealed that the learning-enhanced ADMM exhibited faster convergence compared to the standard ADMM. Further, a decentralized learning framework for predicting consensus and corresponding dual variables was proposed in [15] to warm-start the ADMM algorithm for distributed AC-OPF. This approach notably accelerates the convergence compared to the standard ADMM method.

Alternatively, several studies have employed machine learning models to simplify the nonconvex AC-OPF problem by means of approximation or relaxation techniques. In [141], the authors harnessed a trained NN to enhance the efficiency of the optimality-based bound tightening method, thereby refining variable domains and approaching solutions close to global optima. Utilizing the trained NN, the authors reported an impressive speed improvement of up to six times. In [142], intractable constraints of security- and stability-constrained OPF were transformed into a mixed-integer linear programming problem using a set of NNs. The model achieved cost-optimal solutions that fulfill static and dynamic constraints. A data-driven regression method was proposed in [143] to linearize the nonlinear equations of the OPF problem in the presence of measurement noise. This method transformed the problem into linear-quadratic programming, demonstrating higher accuracy compared to the decoupled linear power flow approach. An ensemble learning approach for convex approximation of single-phase and three-phase AC power flow equations was introduced in [144] and [145]. The methodology leverages polynomial regression as a primary learner and then employs ensemble techniques like gradient boosting and bagging to bolster model performance, ultimately formulating OPF as a convex quadratic programming problem. In terms of both accuracy and computational efficiency, the learning-assisted convex approximation outperformed the conventional semi-definite programming relaxation. Beyond the computational advantages offered by AC-OPF relaxations or approximations, it is essential to note that the results generated by these models may not align with the AC power flow equations, posing challenges for practical scenarios [146]. The authors in both [146] and [147] employed a gradient descent algorithm to restore solution feasibility achieved by simplified or learning-based OPF models, utilizing a state estimation technique. The outcomes demonstrated accuracy enhancements that surpass alternative methods by several orders of magnitude.

5. Challenges and open issues

Research on machine learning applications for the OPF problem has shown promise, yet it remains largely confined to research environments. The shift from a *beneficial* machine-learning model to a *secure* and *widely-adopted* platform is a path yet to be traversed. As the intricate landscape of learning-driven OPF literature and its expansive branches is navigated, it becomes apparent that numerous frontiers remain unaddressed and ripe for further investigation. In this section, the focus is placed on how the power systems research community can spearhead the exploration of these unaddressed territories to pave the way for future research directions in this evolving domain.

5.1. Feasibility–optimality harmony

In addition to achieving accurate predictions, machine learning models face the critical challenge of maintaining solution feasibility with respect to the engineering and physical constraints of the OPF problem. Ensuring that solutions adhere to grid constraints is one of the primary objectives of these models, as it represents the first step toward their practical application. Consequently, this issue has been the central focus of several studies, aiming to guarantee that learning models consistently produce solutions that are viable in real-world scenarios [39,67,89,108,148–150]. Feasibility-guaranteed learning models, such as those described in [148] and [149], offer valuable examples. In [148], feasibility in the predictions of deep NNs for AC-OPF problem was achieved through the use of differentiable implicit layers. Another study [150] explored the factors that impact the robustness of E2E learning predictions in the context of deep NNs and AC-OPF learning. The authors highlighted that while deep NNs excel in predictive accuracy, their limited interpretability hinders drawing definitive conclusions regarding the reliability and practicality of their predictions. For DC-OPF, perspective decision trees were employed in [148] to generate feasible solutions using robust optimization techniques. It was shown that these decision trees not only match deep NNs in predictive performance but

also provide interpretability, potentially making them better suited for real-world and critical scenarios.

Despite concerted efforts to uphold the practicality of these models, a significant path remains to be traversed in order to not only reach *secure* decisions but also to sustain *optimality*. Despite advancements in both E2E and L2O branches, the current models lack a generic framework that ensures absolute feasibility and optimality at the same time. While numerous studies have been able to guarantee complete feasibility (e.g., [39,67,89,108,148–150]), achieving this often comes at the expense of optimality loss. More technically, unconstrained E2E learning models rely on resource-intensive post-processing steps that may not completely adhere to all OPF constraints. Conversely, constrained E2E learning approaches, employing penalty and Lagrangian duality, entail the intricate task of striking a balance between prediction accuracy and constraint satisfaction, a challenge lacking a one-size-fits-all solution and incurring significant computational overhead. The utilization of implicit layers for constraint preservation predominantly guarantees the satisfaction of convex constraints, leaving room for addressing non-convex constraints more effectively. As outlined in Section 4.1, the computational expense of the iterative feasibility test, required to verify the inclusion of all active constraints in active set learning, poses an unresolved challenge, which necessitates further research or alternative approaches. Therefore, the quest for feasibility–optimality fusion continues to be a recurring theme in the field.

5.2. Inter-temporal constraints and discrete variables

Apart from the feasibility–optimality preservation challenge, the second challenge is the inter-temporal constraint compliance. In the domain of OPF, the continuity of solutions across different time steps presents a formidable challenge. Each OPF solution is not an isolated entity but rather intricately connected to its surrounding (past and future) time steps. This challenge becomes more and more important in the face of inter-temporal constraints, including the previous status of units [151], generators' minimum up and down times [151], ramp-up and ramp-down limits along with start-up and shut-down ramp rates [151], energy storage state-of-charge [152] and dynamic line rating constraints [153] as well as demand response schemes [154]. The explored literature confirms that, apart from [84] which introduced an E2E recurrent NN to address the multi-period DC-OPF learning, the majority of existing works dealing with inter-temporal constraints are embedded within reinforcement learning models owing to their inherent capacity in sequential decision-making [104,107,108,110,112]. The practicality of the remaining learning models (E2E and L2O) concerning inter-temporal constraints requires comprehensive investigation. It is a promising avenue to expand the learning models to capture inter-temporal constraints within hours-ahead OPF predictions/optimizations. With an eye to discrete variables inherent in certain intertemporal constraints, another interesting avenue of research lies in the incorporation of discrete elements into physics-informed learning models. This is especially interesting as current models exclusively focus on continuous variables.

5.3. Adaptability

Another challenge arises from the continuous evolution of power systems. It is vital to ensure that the machine-learned models employed for the operation and control of power systems can adjust and maintain their effectiveness over time. Thus, there is an always unaddressed question on the adaptability of learning models facing various network topologies, network expansion, extreme events, different data distributions with different ranges of variations, and interconnection with other energy carriers such as gas [155], liquefied natural gas [156], biomass [157], and water [158]. This holds particular significance because OPF learning models are usually trained on specific datasets that capture distinct system features. Handling datasets for all potential grid scenarios is a substantial challenge due to resource limitations. This suggests why most surveyed works focus on generating high-quality predictions, whether with or without topology awareness. They capture specific input features with limited ranges of variation and types of distribution, often neglecting network expansion, extreme events, and interconnection with other energy carriers. In this context, domain adaptation [159] and transfer learning [160] are active research areas that enable the reutilization of models trained on specific datasets for systems with differing characteristics, reducing the need for exhaustive, resource-intensive dataset creation for each unique scenario. Additionally, exploring ensemble learning [161], which involves combining multiple learning models and implementing automated machine learning [162] techniques designed to automatically optimize various parts of machine learning models, represents another set of potential avenues to enhance the adaptability of machine-learned OPF models.

5.4. Scalability

A persistent challenge in OPF learning is the capacity of learning schemes to effectively manage expanding volumes of data, driven by the growth in power systems' size and the inclusion of security constraints. This challenge, often identified as scalability issues, arises due to the complexities faced by learning models during the training phase. The literature review confirms that training machine learning models for large power systems, characterized by high-dimensional spaces, with massive datasets requires a substantial computational effort, which exacerbates the training process. Taking the E2E AC-OPF learning as an example, the most extensive power system encountered in the literature within a naive unconstrained regression model consisted of a network with 2853 buses, supported by a dataset comprising 10,000 samples [52]. The training process may become computationally expensive due to the high-dimensional nature of expanding datasets. This challenge becomes even more pronounced in constrained regression models, where additional constraints need to be satisfied. Ensuring the scalability of machine learning models for the management of more extensive datasets and intricate constraints continues to be a pivotal factor. There are several strategies to address OPF learning scalability issues, including input feature reduction techniques [163], incremental learning models [164], data augmentation [165], model compression and acceleration [166], as well as distributed machine learning [167]. As an illustration, the capability of compression and distributed learning models to manage systems up to 22,883 (using 8500 samples including security constraints) and 30,000 (utilizing 22,000 samples) buses were testified in [62] and [63], respectively, demonstrating the robustness of the above models in scaling OPF learning models to real-life systems. Nevertheless, additional research is warranted not only for the strategies mentioned above but also for their respective subclasses, including advancements within those fields. Many aspects are in the early stages of exploration, while others remain largely unexplored.

5.5. Standardized datasets, testbeds, and evaluation metrics

The next major challenge pertains to the availability of standardized *datasets* and *testbeds* tailored to various types of E2E and L2O OPF frameworks. The reviewed literature indicates that a vast majority of prior research predominantly depends on synthetic datasets generated within simulation environments. This reliance on synthetic data is largely attributed to the scarcity of real-world datasets, which often stem from privacy and confidentiality limitations. These datasets often incorporate arbitrary variation rates and specific input features (e.g., load profiles, contingency scenarios, and RESs) which results in a lack of consistency and standardization. This issue has propelled *quality OPF data sampling* into the spotlight as an essential area of study, with its primary focus on refining OPF datasets to reflect rare and previously uncovered grid conditions, which are crucial for the success of learning algorithms [72,168–170]. By way of example, a method to enhance training datasets by identifying and integrating cases where NNs fail to meet

constraints was outlined in [72], which ultimately ensures a broader representation of the feature space. Alternatively, a holistic strategy that combines historical significance, broad coverage, and discriminative properties using the Wasserstein distance metric was proposed in [168] to construct more representative datasets. Furthermore, in [169], a chain-rule-driven adversarial sampling framework was suggested to discover local adversarial data points and identify global boundary regions where active constraints shift. Lastly, a bucketized active sampling framework was established in [170] that partitions the input domain into buckets, leverages an independent validation set to isolate regions needing additional data, and selectively acquires new samples to refine OPF training sets.

While systematic dataset generation and benchmarking platforms for OPF learning tasks have been developed in the past [171–173], the absence of unified datasets covering a diverse array of test systems operating under various operating regimes, similar to the real stock data of the S&P 500 [174] in portfolio optimization and ImageNet [175] in computer vision, hinders the ability to rank and compare various learning paradigms against state-of-the-art methods. Recently, researchers at Google DeepMind introduced OPFData [173], a large-scale open-source OPF dataset containing 300,000 solved OPF instances, including N-1 perturbations for grids with up to 13,659 buses. While this dataset lays the foundation for a widely adopted unified resource for researchers, the authors have suggested ways to expand its scope to encompass additional operating conditions.

Moreover, as per the existing literature, a notable disparity exists in the dataset sizes utilized across various OPF learning models. Investigating the necessary data volume (also the quality) for proficiently training diverse learning algorithms represents a significant and continually evolving research area within the field of OPF learning, which demands substantial consideration.

The current state of research often reveals a shortage of standardized testbeds tailored for OPF learning schemes, similar to specialized platforms like Grid2Op [118], initially designed for reinforcement learning benchmarking in power grid control, and Power Grid Lib [176], created for optimizing the OPF problem. The lack of standardized benchmarks for each class of OPF learning models poses a challenge to the research community's ability to benchmark a diverse range of algorithms effectively using common datasets. Establishing such standardized testbeds holds the potential to significantly enhance research endeavors, enabling a more pragmatic application of these models in real-world power system scenarios. Another pressing concern in the intersection of OPF and machine learning is the lack of standardized *evaluation metrics* linked to each subclass of OPF learning models. Consistency in evaluation metrics across different classes of OPF learning models is crucial for fair comparisons, reproducibility, and effective communication of research findings, which ultimately facilitates meaningful advancements in the field.

There are further vivid avenues that the OPF learning problem needs extensive attention, as listed below:

- The application and practicality of many L2O and E2E learning schemes for a broad array of power systems operation problems holding the OPF problem as the building block, including optimal transmission switching, security-constrained unit commitment, trading strategies, etc., require further in-depth exploration and development.
- The current E2E OPF learning models are developed as an empirical risk (neutral) minimization problem, which need to be extended to risk-informed statistical learning concepts to effectively address issues related to outliers and overfitting.
- The intersection of various learning algorithms with decomposition optimization techniques (e.g., dual, optimality condition, and Benders decomposition) commonly leveraged to solve decentralized multi-area OPF merits deeper investigation. In this domain, delving into multi-area OPF using multi-agent learning techniques, where individual agents are responsible for specific electric network regions, demands comprehensive research. Moreover, the application of federated learning [177], aimed at preserving privacy during data-sharing among diverse network areas, opens up another promising avenue for investigation.
- It is widely recognized that the efficacy of reinforcement learning models hinges upon the intricacies of handcrafted reward functions. An interesting avenue for future research lies in exploring inverse reinforcement learning techniques [178] as a sophisticated approach to autonomously derive the reward function for an agent emulating the OPF problem.
- The E2E and L2O frameworks are subject to expanding the OPF problem to involve objectives beyond the least generation cost, such as minimizing transmission loss, voltage deviation, and environmental pollution. To address this multi-objective landscape, multi-objective learning [179] emerges as a promising solution.
- High-voltage DC grids are gaining increased attention due to their efficiency in long-distance transmission and enhanced power flow control. When connected to AC grids, they create a hybrid AC–DC grid, which is more complex to manage [180]. Consequently, the OPF of hybrid AC–DC grids is even more computationally demanding than that of pure AC grids due to the presence of AC/DC converters and the need to coordinate power flow within both the AC and DC grids. It is evident that the development of appropriate machine learning tools for hybrid AC–DC grids is an open challenge that deserves comprehensive examination.

6. Conclusions

This article provided an extensive overview of machine learning applications in the fundamental problem of power systems operation, i.e., the OPF. The commonly developed learning models used for the OPF problem were introduced and classified, followed by a comprehensive comparison of their characteristics across a wide range of factors. Several commonly used evaluation metrics were presented to assess the performance of solutions generated by machine learning models, focusing on OPF learning schemes. Further, valuable perspectives were offered, unresolved challenges were pinpointed, and potential avenues for future research were suggested. Summarizing the key takeaways from this review, several significant lessons emerge. First, unconstrained E2E learning models, despite their ease of implementation, offer predictions of power flow solutions that need reliable post-processing techniques to be applicable in real-world scenarios. Second, constrained E2E learning models, while more complex to implement, offer higher level of feasibility and practicality rates by respecting power flow constraints depending on the chosen architecture, but post-processing procedures may still be necessary to enhance compliance with operational constraints. Thirdly, when it comes to topology awareness, graph-based learning approaches demonstrate exceptional performance, rendering them highly suitable for addressing concerns related to any form of grid contingency. Fourth, constrained reinforcement learning models are highly suitable for managing multi-period OPF while addressing inter-temporal constraints, as they naturally fit well with the concept of sequential decision-making, though requiring customization for specific goals. Finally, it becomes evident that L2O frameworks offer greater practicality and feasibility for real-world implementation by adhering to power flow constraints, although their computational efficiency can be restricted compared to E2E techniques. Consequently, the choice between E2E learning and L2O frameworks may be viewed as a trade-off between computational efficiency and solution feasibility.

CRediT authorship contribution statement

H. Khaloie: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization, Writing – review & editing. **M. Dolányi:** Writing – review & editing, Formal analysis, Data curation. **J.-F. Toubeau:** Writing – review & editing, Formal analysis **F. Vallée:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was carried out in the context of the Energy Transition Fund, DISCRETE project, supported by the FPS Economy, SMEs, Self-Employed and Energy, Belgium.

Data availability

No data were used for the research described in the article.

References

[1] Dommel HW, Tinney WF. Optimal power flow solutions. IEEE Trans Power Appar Syst PAS 1968;87(10):1866–76. https://doi.org/10.1109/TPAS.1968.292150
[2] Wang S, Shahidehpour S, Kirschen D, Mokhtari S, Irisarri G. Short-term generation scheduling with transmission and environmental constraints using an augmented lagrangian relaxation. IEEE Trans Power Syst 1995;10(3):1294–301.
[3] Jabr R, Coonick A, Cory B. A homogeneous linear programming algorithm for the security constrained economic dispatch problem. IEEE Trans Power Syst 2000;15(3):930–36. https://doi.org/10.1109/59.871715
[4] Bouffard F, Galiana F, Conejo A. Market-clearing with stochastic security— part i: formulation. IEEE Trans Power Syst 2005;20(4):1818–26. https://doi.org/10.1109/TPWRS.2005.857016
[5] Singh H, Hao S, Papalexopoulos A. Transmission congestion management in competitive electricity markets. IEEE Trans Power Syst 1998;13(2):672–80. https://doi.org/10.1109/59.667399
[6] Yoshida H, Kawata K, Fukuyama Y, Takayama S, Nakanishi Y. A particle swarm optimization for reactive power and voltage control considering voltage security assessment. IEEE Trans Power Syst 2000;15(4):1232–39. https://doi.org/10.1109/59.898095
[7] Fisher EB, O'Neill RP, Ferris MC. Optimal transmission switching. IEEE Trans Power Syst 2008;23(3):1346–55.
[8] Alguacil N, Motto A, Conejo A. Transmission expansion planning: a mixed-integer lp approach. IEEE Trans Power Syst 2003;18(3):1070–77. https://doi.org/10.1109/TPWRS.2003.814891
[9] Zamzam AS, Baker K. Learning optimal solutions for extremely fast ac optimal power flow. In: 2020 IEEE international conference on communications, control, and computing technologies for smart grids (SmartGridComm); IEEE; 2020. p. 1–6.
[10] Capitanescu F, Martinez Ramos J, Panciatici P, Kirschen D, Marano Marcolini A, Platbrood L, et al. State-of-the-art, challenges, and future trends in security constrained optimal power flow. Electr Power Syst Res 2011;81(8):1731–41. https://doi.org/10.1016/j.epsr.2011.04.003
[11] Capitanescu F. Critical review of recent advances and further developments needed in ac optimal power flow. Electr Power Syst Res 2016;136:57–68. https://doi.org/10.1016/j.epsr.2016.02.008
[12] Molzahn DK, Hiskens IA. A survey of relaxations and approximations of the power flow equations, Found Trends Electr Energy Syst. 2019;4:1–2) 1–221. https://doi.org/10.1561/3100000012
[13] Zhao F, Litvinov E, Zheng T. A marginal equivalent decomposition method and its application to multi-area optimal power flow problems. IEEE Trans Power Syst 2013;29(1):53–61.
[14] Molzahn DK, Dörfler F, Sandberg H, Low SH, Chakrabarti S, Baldick R, et al. A survey of distributed optimization and control algorithms for electric power systems. IEEE Trans Smart Grid 2017;8(6):2941–62.
[15] Mak TW, Chatzos M, Tanneau M, Hentenryck PV. Learning regionally decentralized ac optimal power flows with admm. IEEE Trans Smart Grid 2023;1–1. https://doi.org/10.1109/TSG.2023.3251292
[16] Sun K, Sun XA. A two-level admm algorithm for ac opf with global convergence guarantees. IEEE Trans Power Syst 2021;36(6):5271–81.
[17] Ren Y, Sun Y, Peng M. Deep reinforcement learning based computation offloading in fog enabled industrial internet of things. IEEE Trans Ind Inf 2021;17(7):4978–87. https://doi.org/10.1109/TII.2020.3021024
[18] Xiang X, Gui J, Xiong NN. An integral data gathering framework for supervisory control and data acquisition systems in green iot. IEEE Trans Green Commun Netw 2021;5(2):714–26. https://doi.org/10.1109/TGCN.2021.3068257
[19] Phadke AG, Thorp JS, Phadke AG, Thorp JS. Phasor measurement units and phasor data concentrators. Synchron Phasor Meas Appl 2017;83–109.
[20] Zheng J, Gao DW, Lin L. Smart meters in smart grid: An overview. In: 2013 IEEE green technologies conference (GreenTech); IEEE; 2013. p. 57–64.
[21] Perez J. A guide to digital fault recording event analysis. In: 2010 63rd Annual Conference for Protective Relay Engineers; IEEE; 2010. p. 1–17. https://doi.org/10.1109/CPRE.2010.5469501
[22] Yang T, Zhao L, Li W, Zomaya AY. Reinforcement learning in sustainable energy and electric systems: a survey. Annu Rev Control 2020;49:145–63. https://doi.org/10.1016/j.arcontrol.2020.03.001
[23] Ibrahim MS, Dong W, Yang Q. Machine learning driven smart electric power systems: current trends and new perspectives. Appl Energy 2020;272:115237. https://doi.org/10.1016/j.apenergy.2020.115237
[24] Huang B, Wang J. Applications of physics-informed neural networks in power systems - a review. IEEE Trans Power Syst 2023;38(1):572–88. https://doi.org/10.1109/TPWRS.2022.3162473
[25] Yu L, Qin S, Zhang M, Shen C, Jiang T, Guan X. A review of deep reinforcement learning for smart building energy management. IEEE Internet Things J 2021;8(15):12046–63. https://doi.org/10.1109/JIOT.2021.3078462
[26] Zhu Z, Hu Z, Chan KW, Bu S, Zhou B, Xia S. Reinforcement learning in deregulated energy market: a comprehensive review. Appl Energy 2023;329:120212. https://doi.org/10.1016/j.apenergy.2022.120212
[27] Van Hentenryck P. Machine learning for optimal power flows. Tutor Oper Res Emerg Optim Methods Model Tech Appl, INFORMS 2021;62–82. https://doi.org/10.1287/educ.2021.0234
[28] Hasan F, Kargarian A, Mohammadi A. A survey on applications of machine learning for optimal power flow. In: 2020 IEEE Texas Power and Energy Conference (TPEC); IEEE; 2020. p. 1–6. https://doi.org/10.1109/TPEC48276.2020.9042547
[29] Huang G, Wu F, Guo C. Smart grid dispatch powered by deep learning: a survey. Front Inf Technol Electr Eng 2022;23(5):763–76. https://doi.org/10.1631/FITEE.2000719
[30] Ruan G, Zhong H, Zhang G, He Y, Wang X, Pu T. Review of learning-assisted power system optimization. CSEE J Power Energy Syst 2020;7(2):221–31.
[31] Cunningham P, Cord M, Delany SJ. Supervised learning. In: Machine learning techniques for multimedia: case studies on organization and retrieval. Springer; 2008. pp. 21–49.
[32] Hastie T, Tibshirani R, Friedman J, Hastie T, Tibshirani R, Friedman J. Unsupervised learning, the elements of statistical learning: data mining, inference, and prediction. 2009;485–585.
[33] Sutton RS, Barto AG. Reinforcement learning: an introduction. MIT press; 2018.
[34] Kotary J, Fioretto F, Van Hentenryck P, Wilder B. End-to-end constrained optimization learning: a survey, arXiv preprint 2103.16378 (2021).
[35] Xia F, Sun K, Yu S, Aziz A, Wan L, Pan S, et al. Graph learning: a survey. IEEE Trans Artif Intell 2021;2(2):109–27.
[36] Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, et al. Generative adversarial nets. Adv Neural Inf Process Syst 27 2014.
[37] Dener A, Miller MA, Churchill RM, Munson T, Chang C-S. Training neural networks under physical constraints using a stochastic augmented lagrangian approach, arXiv preprint 2009.07330 (2020).
[38] Stooke A, Achiam J, Abbeel P. Responsive safety in reinforcement learning by pid lagrangian methods. In: International Conference on Machine Learning, PMLR; 2020. p. 9133–43.
[39] Donti PL, Rolnick D, Kolter JZ. Dc3: a learning method for optimization with hard constraints, arXiv preprint 2104.12225 (2021).
[40] Crozier C, Baker K. Data-driven probabilistic constraint elimination for accelerated optimal power flow. In: 2022 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2022. p. 1–5.
[41] Sun S, Cao Z, Zhu H, Zhao J. A survey of optimization methods from a machine learning perspective. IEEE Trans Cybern 2019;50(8):3668–81.
[42] Panesar A. Machine learning and AI for healthcare. Springer; 2021. https://doi.org/10.1007/978-1-4842-6537-6
[43] Saskia. How to normalize the rmse; 2019. https://www.marinedatascience.co/blog/2019/01/07/normalizing-the-rmse/
[44] Li Y, Zhao C, Liu C. Solving non-linear optimization problem in engineering by model-informed generative adversarial network (mi-gan). In: 2022 IEEE International Conference on Data Mining Workshops (ICDMW); 2022. p. 198–205. https://doi.org/10.1109/ICDMW58026.2022.00035
[45] Li Y, Zhao C, Liu C. Model-informed generative adversarial network (mi-gan) for learning optimal power flow, arXiv preprint 2206.01864 (2022).
[46] Wang J, Srikantha P. Fast optimal power flow with guarantees via an unsupervised generative model. IEEE Trans Power Syst 2022.
[47] Canyasse R, Dalal G, Mannor S. Supervised learning for optimal power flow as a real-time proxy. In: 2017 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT); IEEE; 2017. p. 1–5.
[48] Sun Y, Fan X, Huang Q, Li X, Huang R, Yin T, et al. Local feature sufficiency exploration for predicting security-constrained generation dispatch in multi-area power systems. In: 2018 17th IEEE International Conference on Machine Learning and Applications (ICMLA); IEEE; 2018. p. 1283–89. https://doi.org/10.1109/ICMLA.2018.00208

[49] Yang K, Gao W, Fan R. Optimal power flow estimation using one-dimensional convolutional neural network. In: 2021 North American Power Symposium (NAPS); IEEE; 2021. p. 1–6. https://doi.org/10.1109/NAPS52732.2021.9654438

[50] Mak TW, Fioretto F, VanHentenryck P. Load encoding for learning ac-opf, arXiv preprint 2101.03973 (2021).

[51] Wu H, Xu Z. Fast dc optimal power flow based on deep convolutional neural network. In: 2022 IEEE 5th International Electrical and Energy Conference (CIEEC); IEEE; 2022. p. 2508–12. https://doi.org/10.1109/CIEEC54735.2022.9846143

[52] Falconer T, Mones L. Leveraging power grid topology in machine learning assisted optimal power flow. IEEE Trans Power Syst 2022;1–13. https://doi.org/10.1109/TPWRS.2022.3187218

[53] Dong Z, Hou K, Liu Z, Yu X, Jia H, Zhang C. A sample-efficient opf learning method based on annealing knowledge distillation. IEEE Acces 2022;10:99724–33. https://doi.org/10.1109/ACCESS.2022.3207146

[54] Lotfi A, Pirnia M. Constraint-guided deep neural network for solving optimal power flow. Electr Power Syst Res 2022;211:108353. https://doi.org/10.1016/j.epsr.2022.108353

[55] Chen Y, Lakshminarayana S, Maple C, Poor HV. A meta-learning approach to the optimal power flow problem under topology reconfigurations. IEEE Open Access J Power Energy 2022;9:109–20. https://doi.org/10.1109/OAJPE.2022.3140314

[56] Jia Y, Bai X, Zheng L, Weng Z, Li Y. Convopf-dop: a data-driven method for solving ac-opf based on cnn considering different operation patterns. IEEE Trans Power Syst 2022;38(1):853–60.

[57] Rahman J, Feng C, Zhang J. A learning-augmented approach for ac optimal power flow. Int J Electr Power Energy Syst 2021;130:106908.

[58] Rahman J, Feng C, Zhang J. Machine learning-aided security constrained optimal power flow. In: 2020 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2020. p. 1–5. https://doi.org/10.1109/PESGM41954.2020.9281941

[59] Zhou M, Chen M, Low SH. Deepopf-ft: one deep neural network for multiple ac-opf problems with flexible topology. IEEE Trans Power Syst 2022;38(1):964–67.

[60] Huang W, Pan X, Chen M, Low SH. Deepopf-v: Solving ac-opf problems efficiently. IEEE Trans Power Syst 2022;37(1):800–03. https://doi.org/10.1109/TPWRS.2021.3114092

[61] Pan X, Huang W, Chen M, Low SH. Deepopf-al: augmented learning for solving ac-opf problems with a multi-valued load-solution mapping. In: Proceedings of the 14th ACM International Conference on Future Energy Systems; 2023. p. 42–47.

[62] Vyakaranam B, Mahapatra K, Li X, Wang H, Etingov P, Hou Z, et al. Novel data-driven distributed learning framework for solving ac power flow for large interconnected systems. IEEE Open Access J Power Energy 2021;8:281–92.

[63] Park S, Chen W, Mak TW, Van Hentenryck P. Compact optimization learning for ac optimal power flow. IEEE Trans Power Syst 2023.

[64] Klamkin M, Tanneau M, Mak TWK, Van Hentenryck P. Active bucketized learning for acopf optimization proxies, arXiv preprint 2208.07497 (2022).

[65] Zhai Y, Ye H, Zhang L, Ge Y. Model information-aided deep learning for corrective ac power flow. In: 2022 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2022. p. 1–5. https://doi.org/10.1109/PESGM48719.2022.9916916

[66] Zhao T, Pan X, Chen M, Venzke A, Low SH. Deepopf+: a deep neural network approach for dc optimal power flow for ensuring feasibility. In: 2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm); IEEE; 2020. p. 1–6.

[67] Pan X, Zhao T, Chen M, Zhang S. Deepopf: a deep neural network approach for security-constrained dc optimal power flow. IEEE Trans Power Syst 2020;36(3):1725–35.

[68] Pan X. Deepopf: deep neural networks for optimal power flow. In: Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation; 2021. p. 250–51.

[69] Zhao T, Pan X, Chen M, Low SH. Ensuring dnn solution feasibility for optimization problems with convex constraints and its application to dc optimal power flow problems, arXiv preprint 2112.08091 (2021).

[70] Wang Z, Menke J-H, Schäfer F, Braun M, Scheidler A. Approximating multi-purpose ac optimal power flow with reinforcement trained artificial neural network. Energy AI 2022;7:100133.

[71] Guha N, Wang Z, Wytock M, Majumdar A. Machine learning for ac optimal power flow, arXiv preprint 1910.08842 (2019).

[72] Hu Z, Zhang H. Optimal power flow based on physical-model-integrated neural network with worth-learning data generation, arXiv preprint 2301.03766 (2023).

[73] Nellikkath R, Chatzivasileiadis S. Physics-informed neural networks for minimising worst-case violations in dc optimal power flow. In: 2021 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm); 2021. p. 419–24. https://doi.org/10.1109/SmartGridComm51999.2021.9632308

[74] Nellikkath R, Chatzivasileiadis S. Physics-informed neural networks for ac optimal power flow. Electr Power Syst Res 2022;212:108412.

[75] Nellikkath R, Chatzivasileiadis S. Minimizing worst-case violations of neural networks, arXiv preprint 2212.10930 (2022).

[76] Venzke A, Qu G, Low S, Chatzivasileiadis S. Learning optimal power flow: worst-case guarantees for neural networks. In: 2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm); IEEE; 2020. p. 1–7.

[77] Chen K, Bose S, Zhang Y. Unsupervised deep learning for ac optimal power flow via lagrangian duality. In: GLOBECOM 2022-2022 IEEE Global Communications Conference; IEEE; 2022. p. 5305–10.

[78] Fioretto F, Mak TWK, Van Hentenryck P. Predicting ac optimal power flows: combining deep learning and lagrangian dual methods. In: Proceedings of the AAAI conference on artificial intelligence; Vol. 34. 2020. p. 630–37.

[79] Fioretto F, Van Hentenryck P, Mak TW, Tran C, Baldo F, Lombardi M. Lagrangian duality for constrained deep learning. In: Machine Learning and Knowledge Discovery in Databases. Applied Data Science and Demo Track: European Conference, ECML PKDD 2020; Vol. Proceedings, Part V. Ghent, Belgium: Springer; 2021. p. 118–35. [September 14–18, 2020].

[80] Chatzos M, Fioretto F, Mak TWK, Van Hentenryck P. High-fidelity machine learning approximations of large-scale optimal power flow, arXiv preprint 2006.16356 (2020).

[81] Mak TW, Fioretto F, Van Hentenryck P. Load embeddings for scalable{AC-OPF} learning, ArXivorg. 2023.

[82] Chatzos M, Mak TWK, Hentenryck PV. Spatial network decomposition for fast and scalable ac-opf learning. IEEE Trans Power Syst 2022;37(4):2601–12. https://doi.org/10.1109/TPWRS.2021.3124726

[83] Dinh MH, Fioretto F, Mohammadian M, Baker K. Towards understanding the unreasonable effectiveness of learning ac-opf solutions, arXiv preprint 2111.11168 (2021).

[84] Mohammadian M, Baker K, Dinh MH, Fioretto F. Learning solutions for intertemporal power systems optimization with recurrent neural networks. In: 2022 17th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS); IEEE; 2022. p. 1–6. https://doi.org/10.1109/PMAPS53380.2022.9810638

[85] Kotary J, Fioretto F, Van Hentenryck P. Learning hard optimization problems: a data generation perspective. Adv Neural Inf Process Syst 2021;34:24981–92.

[86] Park S, Van Hentenryck P. Self-supervised primal-dual learning for constrained optimization. In: Proceedings of the AAAI Conference on Artificial Intelligence; Vol. 37. 2023. p. 4052–60.

[87] Chen G, Qin J. On the choice of loss function in learning-based optimal power flow, arXiv preprint 2402.00773 (2024).

[88] Kim M, Kim H. Projection-aware deep neural network for dc optimal power flow without constraint violations. In: 2022 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm); IEEE; 2022. p. 116–21. https://doi.org/10.1109/SmartGridComm52983.2022.9961047

[89] Li M, Kolouri S, Mohammadi J. Learning to solve optimization problems with hard linear constraints. IEEE Acces 2023;11:59995–60004.

[90] Zhang L, Tabas D, Zhang B. An efficient learning-based solver for two-stage dc optimal power flow with feasibility guarantees. 2023.

[91] Kolter MZ, Duvenaud D, Johnson M. Deep implicit layers-neural odes, deep equilibirum models, and beyond, 2020, NeurIPS tutorial. 2020.

[92] Amos B, Kolter JZ. Optnet: differentiable optimization as a layer in neural networks. In: International Conference on Machine Learning, PMLR; 2017. p. 136–45.

[93] Agrawal A, Amos B, Barratt S, Boyd S, Diamond S, Kolter JZ. Differentiable convex optimization layers. Adv Neural Inf Process Syst 2019;32.

[94] Chen B, Donti PL, Baker K, Kolter JZ, Bergés M. Enforcing policy feasibility constraints through differentiable projection for energy optimization. In: Proceedings of the Twelfth ACM International Conference on Future Energy Systems; 2021. p. 199–210.

[95] Xia F, Sun K, Yu S, Aziz A, Wan L, Pan S, et al. Graph learning: a survey. IEEE Trans Artif Intell 2021;2(2):109–27.

[96] Hansen JB, Anfinsen SN, Bianchi FM. Power flow balancing with decentralized graph neural networks. In: IEEE Transactions on Power Systems; 2022.

[97] Liu S, Wu C, Zhu H. Topology-aware graph neural networks for learning feasible and adaptive ac-opf solutions. In: IEEE Transactions on Power Systems; 2022.

[98] Liu S, Wu C, Zhu H. Graph neural networks for learning real-time prices in electricity market, arXiv preprint 2106.10529 (2021).

[99] Gao M, Yu J, Yang Z, Zhao J. A physics-guided graph convolution neural network for optimal power flow. In: IEEE Transactions on Power Systems; 2023.

[100] Yang M, Liu J, Qiu G, Liu T, Z. Tang, Ac optimal power flows: combining lagrangian dual and physics-guided graph neutral networks, Authorea Preprints. 2023.

[101] Brockman G, Cheung V, Pettersson L, Schneider J, Schulman J, Tang J, et al. Openai gym, arXiv preprint 1606.01540 (2016).

[102] Wolgast T, Nieße A. Learning the optimal power flow: environment design matters, arXiv preprint 2403.17831 (2024).

[103] Nie H, Chen Y, Song Y, Huang S. A general real-time opf algorithm using ddpg with multiple simulation platforms. 2019 IEEE innovative smart grid technologies-asia (ISGT asia). IEEE; 2019. p. 3713–18.

[104] Yan Z, Xu Y. Real-time optimal power flow: a Lagrangian based deep reinforcement learning approach. IEEE Trans Power Syst 2020;35(4):3270–73.

[105] Xiao T, Chen Y, Diao H, Huang S, Shen C. On fast-converged reinforcement learning for optimal dispatch of large-scale power systems under transient security constraints, arXiv preprint 2304.08320 (2023).

[106] Zhen H, Zhai H, Ma W, Zhao L, Weng Y, Xu Y, et al. Design and tests of reinforcement-learning-based optimal power flow solution generator. Energy Rep 2022;8:43–50.

[107] Woo JH, Wu L, Park J-B, Roh JH. Real-time optimal power flow using twin delayed deep deterministic policy gradient algorithm. IEEE Acces 2020;8:213611–18. https://doi.org/10.1109/ACCESS.2020.3041007

[108] Wu T, Scaglione A, Arnold D. Constrained reinforcement learning for stochastic dynamic optimal power flow control. In: 2023 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2023. p. 1–5.

[109] Han X, Mu C, Yan J, Niu Z. An autonomous control technology based on deep reinforcement learning for optimal active power dispatch. Int J Electr Power Energy Syst 2023;145:108686.

[110] Sayed AR, Wang C, Anis H, Bi T. Feasibility constrained online calculation for real-time optimal power flow: a convex constrained deep reinforcement learning approach. In: IEEE Transactions on Power Systems; 2022.
[111] Wang T, Tang Y. Transfer-reinforcement-learning-based rescheduling of differential power grids considering security constraints. Appl Energy 2022;306:118121.
[112] Sayed AR, Zhang X, Wang G, Wang C, Qiu J. Optimal operable power flow: sample-efficient holomorphic embedding-based reinforcement learning. In: IEEE Transactions on Power Systems; 2023. p. 1–13. https://doi.org/10.1109/TPWRS.2023.3266773
[113] Huang Z, Yu T, Pan Z, Deng B, Zhang X, Wu Y, et al. Stochastic dynamic power dispatch with high generalization and few-shot adaption via contextual meta graph reinforcement learning. Int J Electr Power Energy Syst 2024;162:110272.
[114] Zhou Y, Zhang B, Xu C, Lan T, Diao R, Shi D, et al. Lee, a data-driven method for fast ac optimal power flow solutions via deep reinforcement learning. J Modern Power Syst Clean Energy 2020;8(6):1128–39.
[115] Zhou Y, Lee W, Diao R, Shi D. Deep reinforcement learning based real-time ac optimal power flow considering uncertainties. J Modern Power Syst Clean Energy 2021;10(5):1098–109.
[116] López-Cardona Á, Bernárdez G, Barlet-Ros P, Cabellos-Aparicio A. Proximal policy optimization with graph neural networks for optimal power flow, arXiv preprint 2212.12470 (2022).
[117] Zhao Y, Liu J, Liu X, Yuan K, Ren K, Yang M. A graph-based deep reinforcement learning framework for autonomous power dispatch on power systems with changing topologies. In: 2022 IEEE Sustainable Power and Energy Conference (iSPEC); IEEE; 2022. p. 1–5. https://doi.org/10.1109/iSPEC54162.2022.10033001
[118] Donnot B. Grid2op- a testbed platform to model sequential decision making in power systems, GitHub repository. 2020.
[119] Misra S, Roald L, Ng Y. Learning for constrained optimization: identifying optimal active constraint sets. INFORMS J Comput 2022;34(1):463–80. https://doi.org/10.1287/ijoc.2020.1037
[120] Robson A, Jamei M, Ududec C, Mones L. Learning an optimally reduced formulation of opf through meta-optimization, arXiv preprint 1911.06784 (2019).
[121] Deka D, Misra S. Learning for dc-opf: classifying active sets using neural nets. In: 2019 IEEE Milan PowerTech; IEEE; 2019. p. 1–6.
[122] Wu H, Wang M, Xu Z, Jia Y. Active constraint identification assisted dc optimal power flow. In: 2022 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia); IEEE; 2022. p. 185–89. https://doi.org/10.1109/ICPSAsia55496.2022.9949655
[123] Phamh T, Li X. Constraints and variables reduction for optimal power flow using hierarchical graph neural networks with virtual node-splitting, arXiv preprint 2411.06268 (2024).
[124] Khazaei H, Zhao Y. Physics-aware fast learning and inference for predicting active set of dc-opf. In: 2022 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT); IEEE; 2022. p. 1–5.
[125] Liu S, Guo Y, Tang W, Sun H, Huang W, Hou J. Varying condition scopf optimization based on deep learning and knowledge graph. In: IEEE Transactions on Power Systems; 2022.
[126] Liu S, Guo Y, Tang W, Sun H, Huang W. Predicting active constraints set in security-constrained optimal power flow via deep neural network. In: 2021 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2021. p. 01–05. https://doi.org/10.1109/PESGM46819.2021.9637964
[127] Hasan F, Kargarian A, Mohammadi J. Hybrid learning aided inactive constraints filtering algorithm to enhance ac opf solution time. IEEE Trans Ind Appl 2021;57(2):1325–34.
[128] Chen Y, Zhang L, Zhang B. Learning to solve dcopf: a duality approach. Electr Power Syst Res 2022;213:108595.
[129] Bose S, Chen K, Zhang Y. A mixture-of-gradient-experts framework for accelerating ac optimal power flow; 2024). https://arxiv.org/abs/2312.07276
[130] Ferrando R, Pagnier L, Mieth R, Liang Z, Dvorkin Y, Bienstock D, et al. Physics-informed machine learning for electricity markets: a nyiso case study. In: IEEE Transactions on Energy Markets, Policy and Regulation; 2023.
[131] He X, Wen H, Zhang Y, Chen Y. Enabling fast unit commitment constraint screening via learning cost model; 2022). https://arxiv.org/abs/2212.00483
[132] Jiménez-Cordero A, Morales JM, Pineda S. Warm-starting constraint generation for mixed-integer optimization: a machine learning approach. Knowl-Based Syst 2022;253:109570. https://doi.org/10.1016/j.knosys.2022.109570. https://www.sciencedirect.com/science/article/pii/S0950705122007894
[133] Pineda S, Morales JM, Jiménez-Cordero A. Data-driven screening of network constraints for unit commitment. IEEE Trans Power Syst 2020;35(5):3695–705. https://doi.org/10.1109/TPWRS.2020.2980212
[134] Diehl F. Warm-starting ac optimal power flow with graph neural networks. In: 33rd Conference on Neural Information Processing Systems (NeurIPS 2019); 2019. p. 1–6.
[135] Baker K. Learning warm-start points for ac optimal power flow. In: 2019 IEEE 29th International Workshop on Machine Learning for Signal Processing (MLSP); IEEE; 2019. p. 1–6.
[136] Sadat SA, Sahraei-Ardakani M. Initializing successive linear programming solver for acopf using machine learning. In: 2020 52nd North American Power Symposium (NAPS); IEEE; 2021. p. 1–6. https://doi.org/10.1109/NAPS50074.2021.9449706
[137] Dong W, Xie Z, Kestor G, Li D. Smart-pgsim: using neural network to accelerate ac-opf power grid simulation. In: SC20: International Conference for High Performance Computing, Networking, Storage and Analysis; IEEE; 2020. p. 1–15. https://doi.org/10.1109/SC41405.2020.00067
[138] Zhang L, Zhang B. Learning to solve the ac optimal power flow via a lagrangian approach. In: 2022 North American Power Symposium (NAPS); IEEE; 2022. p. 1–6. https://doi.org/10.1109/NAPS56150.2022.10012237
[139] Zeng S, Kody A, Kim Y, Kim K, Molzahn DK. A reinforcement learning approach to parameter selection for distributed optimal power flow. Electr Power Syst Res 2022;212:108546.
[140] Biagioni D, Graf P, Zhang X, Zamzam AS, Baker K, King J. Learning-accelerated admm for distributed dc optimal power flow. IEEE Control Syst Lett 2020;6:1–6.
[141] Cengil F, Nagarajan H, Bent R, Eksioglu S, Eksioglu B. Learning to accelerate globally optimal solutions to the ac optimal power flow problem. Electr Power Syst Res 2022;212:108275.
[142] Murzakhanov I, Venzke A, Misyris GS, Chatzivasileiadis S. Neural networks for encoding dynamic security-constrained optimal power flow, arXiv preprint 2003.07939 (2020).
[143] Liu Y, Wang Y, Zhang N, Lu D, Kang C. A data-driven approach to linearize power flow equations considering measurement noise. IEEE Trans Smart Grid 2020;11(3):2576–87. https://doi.org/10.1109/TSG.2019.2957799
[144] Hu R, Li Q, Lei S. Ensemble learning based linear power flow. In: 2020 IEEE Power & Energy Society General Meeting (PESGM); IEEE; 2020. p. 1–5. https://doi.org/10.1109/PESGM41954.2020.9281793
[145] Hu R, Li Q, Qiu F. Ensemble learning based convex approximation of three-phase power flow. IEEE Trans Power Syst 2021;36(5):4042–51. https://doi.org/10.1109/TPWRS.2021.3055481
[146] Taheri B, Molzahn DK. Ac power flow feasibility restoration via a state estimation-based post-processing algorithm. Electr Power Syst Res 2024;235:110642.
[147] Taheri B, Molzahn DK. Restoring ac power flow feasibility from relaxed and approximated optimal power flow models. In: 2023 American Control Conference (ACC); IEEE; 2023. p. 4463–70.
[148] Han J, Wang W, Yang C, Niu M, Yang C, Yan L, et al. Frmnet: a feasibility restoration mapping deep neural network for ac optimal power flow. In: IEEE Transactions on Power Systems; 2024.
[149] Stratigakos A, Pineda S, Morales JM, Kariniotakis G. Interpretable machine learning for dc optimal power flow with feasibility guarantees. In: IEEE Transactions on Power Systems; 2023.
[150] Dinh MH, Fioretto F, Mohammadian M, Baker K. An analysis of the reliability of ac optimal power flow deep learning proxies. In: 2023 IEEE PES Innovative Smart Grid Technologies Latin America (ISGT-LA); IEEE; 2023. p. 170–74.
[151] Pineda S, Fernández-Blanco R, Morales JM. Time-adaptive unit commitment. IEEE Trans Power Syst 2019;34(5):3869–78. https://doi.org/10.1109/TPWRS.2019.2903486
[152] Khaloie H, Faraji J, Vallée F, Lai CS, Toubeau J-F, Lai LL. Risk-aware battery bidding with a novel benchmark selection under second-order stochastic dominance. IEEE Trans Ind Appl 2023;59(3):3009–18. https://doi.org/10.1109/TIA.2023.3245047
[153] Nick M, Alizadeh-Mousavi O, Cherkaoui R, Paolone M. Security constrained unit commitment with dynamic thermal line rating. IEEE Trans Power Syst 2016;31(3):2014–25. https://doi.org/10.1109/TPWRS.2015.2445826
[154] Zhao C, Wang J, Watson J-P, Guan Y. Multi-stage robust unit commitment considering wind and demand response uncertainties. IEEE Trans Power Syst 2013;28(3):2708–17.
[155] Constante-Flores GE, Conejo AJ, Qiu F. Daily scheduling of generating units with natural-gas market constraints. Eur J Oper Res 2024;313(1):387–99.
[156] Khaloie H, Vallée F. Day-ahead dispatch of liquid air energy storage coupled with lng regasification in electricity and lng markets. In: IEEE Transactions on Power Systems; 2023. p. 1–14. https://doi.org/10.1109/TPWRS.2023.3324150
[157] Khaloie H, Toubeau J-F, Vallée F, Lai CS, Lai LL. An innovative coalitional trading model for a biomass power plant paired with green energy resources. IEEE Trans Sustain Energy 2022;13(2):892–904. https://doi.org/10.1109/TSTE.2021.3138777
[158] Oikonomou K, Parvania M. Optimal coordination of water distribution energy flexibility with power systems operation. IEEE Trans Smart Grid 2019;10(1):1101–10. https://doi.org/10.1109/TSG.2018.2824308
[159] Farahani A, Voghoei S, Rasheed K, Arabnia HR. A brief review of domain adaptation. In: Advances in data science and information engineering: proceedings from ICDATA 2020 and IKE 2020; 2021. p. 877–94.
[160] Zhuang F, Qi Z, Duan K, Xi D, Zhu Y, Zhu H, et al. A comprehensive survey on transfer learning. In: Proceedings of the IEEE; Vol. 109. 2020. p. 43–76 (1.
[161] Ganaie MA, Hu M, Malik A, Tanveer M, Suganthan P. Ensemble deep learning: A review. Eng Appl Artif Intel 2022;115:105151.
[162] Zöller M-A, Huber MF. Benchmark and survey of automated machine learning frameworks. J Artif Intell Res 2021;70:409–72.
[163] Jia W, Sun M, Lian J, Hou S. Feature dimensionality reduction: a review, complex & intelligent Systems. 2022;8(3):2663–93.
[164] Castro FM, Marín-Jiménez MJ, Guil N, Schmid C, Alahari K. End-to-end incremental learning. In: Proceedings of the European conference on computer vision (ECCV); 2018. p. 233–48.
[165] Mumuni A, Mumuni F. Data augmentation: a comprehensive survey of modern approaches, array. 2022;100258.
[166] Deng L, Li G, Han S, Shi L, Xie Y. Model compression and hardware acceleration for neural networks: a comprehensive survey. In: Proceedings of the IEEE; Vol. 108. 2020. p. 485–532 (4.
[167] Verbraeken J, Wolting M, Katzy J, Kloppenburg J, Verbelen T, Rellermeyer JS. A survey on distributed machine learning. ACM Comput Surv (CSUR) 2020;53(2):1–33.
[168] Bugaje A-AB, Cremer JL, Strbac G. Generating quality datasets for real-time security assessment: balancing historically relevant and rare feasible operating conditions. Int J Electr Power Energy Syst 2023;154:109427.
[169] Gao M, Yu J, Yang Z. Improving the robustness of data-driven opf by nonlinearity-focused sampling. In: IEEE Transactions on Power Systems; 2024.

[170] Klamkin M, Tanneau M, Mak TW, Van Hentenryck P. Bucketized active sampling for learning acopf. Electr Power Syst Res 2024;235:110697.
[171] Joswig-Jones T, Baker K, Zamzam AS. Opf-learn: an open-source framework for creating representative ac optimal power flow datasets. In: 2022 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT); IEEE; 2022. p. 1–5.
[172] Nadal IV, Chevalier S. Scalable bilevel optimization for generating maximally representative opf datasets. In: 2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE); IEEE; 2023. p. 1–6.
[173] Lovett S, Zgubic M, Liguori S, Madjiheurem S, Tomlinson H, Elster S, et al. Opfdata: large-scale datasets for ac optimal power flow with topological perturbations, arXiv preprint 2406.07234 (2024).
[174] S&p 500® | S&p Dow Jones indices. https://www.spglobal.com/spdji/en/indices/equity/sp-500/#overview
[175] Imagenet. https://www.image-net.org/
[176] Babaeinejadsarookolaee S, Birchfield A, Christie RD, Coffrin C, DeMarco C, Diao R, et al. The power grid library for benchmarking ac optimal power flow algorithms, arXiv preprint 1908.02788 (2019).
[177] Li L, Fan Y, Tse M, Lin K-Y. A review of applications in federated learning. Comput Ind Eng 2020;149:106854.
[178] Arora S, Doshi P. A survey of inverse reinforcement learning: challenges, methods and progress. Artif Intell 2021;297:103500.
[179] Van Moffaert K, Nowé A. Multi-objective reinforcement learning using sets of pareto dominating policies. J Mach Learn Res 2014;15(1):3483–512.
[180] Ergun H, Dave J, Van Hertem D, Geth F. Optimal power flow for ac–dc grids: formulation, convex relaxation, linear approximation, and implementation. IEEE Trans Power Syst 2019;34(4):2980–90. https://doi.org/10.1109/TPWRS.2019.2897835